Recursive Formulae for Trotter-Suzuki Decompositions

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Introduction

In this document we gather the results of our derivations within the Mathematica notebook (Efficient_Trotterizations.nb). First we define our basis for each order n, which is written in terms of right-nested commutators as described by Arnal *et al.* (2021). We also symmetrize the basis according to the symmetry $A \leftrightarrow B$, which allows us to derive the recursive formulae for one of the steps only (we choose the step with the A operator). For orders $n \geq 6$ there also exist identities, which help reduce the total number of basis commutators. We also write down these relations.

In the second part we present the recursive formulae for the scheme coefficients $\alpha_A^{(i,i-1)}$, $\beta_A^{(i,i-1)}$, $\gamma_{j,A}^{(i,i-1)}$, ... These are written down in terms of the step done with the commutator A. By replacing the parameters a_i with b_i and the scheme coefficients with their respective counterparts, it is possible to find the formulae for the step in the B operator. In this formulation the coefficients ν and σ simply add up the scheme parameters as such:

$$\nu^{(i)} = \nu^{(i-1)} + a_i, \tag{1}$$

$$\sigma^{(i-1)} = \sigma^{(i-2)} + b_i. \tag{2}$$

Basis

2nd order

$$C_1 = \left[A^2 B\right], \quad \tilde{\alpha} = -\frac{1}{24},$$
$$C_2 = \left[B^2 A\right], \quad \tilde{\beta} = \frac{1}{12}.$$

4th order

$$D_1 = \left[A^4 B\right] = \left[A^2 C_1\right], \qquad \tilde{\gamma}_1 = \frac{7}{5760},$$
$$D_2 = \left[A^2 BAB\right] = -\left[A^2 C_2\right] = \left[ABA^2 B\right] = \left[ABC_1\right], \quad \tilde{\gamma}_2 = \frac{1}{480},$$
$$D_3 = \left[BA^3 B\right] = \left[BAC_1\right], \qquad \tilde{\gamma}_3 = \frac{1}{360},$$
$$D_4 = \left[AB^3 A\right] = \left[ABC_2\right], \qquad \tilde{\gamma}_4 = \frac{1}{360},$$
$$D_5 = \left[B^2 ABA\right] = -\left[B^2 C_1\right] = \left[BAB^2 A\right] = \left[BAC_2\right], \quad \tilde{\gamma}_5 = -\frac{1}{120},$$
$$D_6 = \left[B^4 A\right] = \left[B^2 C_2\right], \qquad \tilde{\gamma}_6 = -\frac{1}{720}.$$

6th order

$$E_1 = \left[A^6 B\right] = \left[A^2 D_1\right], \qquad \tilde{\delta}_1 = -\frac{31}{967680},$$
$$E_2 = \left[A^2 BA^3 B\right] = \left[A^2 D_3\right], \qquad \tilde{\delta}_2 = -\frac{19}{80640},$$
$$E_3 = \left[ABA^4 B\right] = \left[ABD_1\right], \qquad \tilde{\delta}_3 = \frac{23}{161280},$$
$$E_4 = \left[BA^5 B\right] = \left[BAD_1\right], \qquad \tilde{\delta}_4 = -\frac{1}{10080},$$
$$E_5 = \left[A^2 BABAB\right] = \left[A^2 B^2 A^2 B\right] = -\left[A^2 D_5\right], \quad \tilde{\delta}_5 = -\frac{1}{10080},$$
$$E_6 = \left[ABA^2 BAB\right] = \left[ABABA^2 B\right] = \left[ABD_2\right], \quad \tilde{\delta}_6 = -\frac{1}{20160},$$
$$E_7 = \left[AB^2 A^3 B\right] = \left[ABD_3\right], \qquad \tilde{\delta}_7 = \frac{13}{60480},$$
$$E_8 = \left[BABA^3 B\right] = \left[BAD_3\right], \qquad \tilde{\delta}_8 = -\frac{1}{5040},$$
$$E_9 = \left[BA^3 BAB\right] = \left[BA^2 BA^2 B\right] = \left[BAD_2\right], \qquad \tilde{\delta}_9 = -\frac{1}{3360},$$

$$
\begin{aligned}
E_{10} &= \left[AB^3ABA\right] = \left[AB^2AB^2A\right] = \left[ABD_5\right], & \tilde{\delta}_{10} &= -\frac{19}{40320},\\
E_{11} &= \left[ABAB^3A\right] = \left[ABD_4\right], & \tilde{\delta}_{11} &= \frac{1}{6720},\\
E_{12} &= \left[BA^2B^3A\right] = \left[BAD_4\right], & \tilde{\delta}_{12} &= -\frac{1}{7560},\\
E_{13} &= \left[BAB^2ABA\right] = \left[BABAB^2A\right] = \left[BAD_5\right], & \tilde{\delta}_{13} &= \frac{1}{1008},\\
E_{14} &= \left[B^2ABABA\right] = \left[B^2A^2B^2A\right] = -\left[B^2D_2\right], & \tilde{\delta}_{14} &= -\frac{1}{10080},\\
E_{15} &= \left[AB^5A\right] = \left[ABD_6\right], & \tilde{\delta}_{15} &= -\frac{1}{10080},\\
E_{16} &= \left[BAB^4A\right] = \left[BAD_6\right], & \tilde{\delta}_{16} &= \frac{1}{10080},\\
E_{17} &= \left[B^2AB^3A\right] = \left[B^2D_4\right], & \tilde{\delta}_{17} &= \frac{1}{5040},\\
E_{18} &= \left[B^6A\right] = \left[B^2D_6\right], & \tilde{\delta}_{18} &= \frac{1}{30240}.
\end{aligned}
$$

$$
\begin{aligned}
\left[A^4BAB\right] = \left[A^3BA^2B\right] = \left[A^2D_2\right] &= 2E_2 - E_3,\\
\left[A^3B^2AB\right] = -\left[A^2D_4\right] &= 3E_5 - 3E_6 + E_7,\\
\left[B^2A^4B\right] = \left[B^2D_1\right] &= 2E_8 - E_9,\\
\left[A^2B^4A\right] = \left[A^2D_6\right] &= 2E_{11} - E_{10},\\
\left[B^3A^2BA\right] = -\left[B^2D_3\right] &= 3E_{14} - 3E_{13} + E_{12},\\
\left[B^4ABA\right] = \left[B^3AB^2A\right] = \left[B^2D_5\right] &= 2E_{17} - E_{16}.
\end{aligned}
$$

8th order

$$
\begin{aligned}
F_1 &= \left[A^8B\right] = \left[A^2E_1\right], & \tilde{\epsilon}_1 &= \frac{127}{154828800},\\
F_2 &= \left[A^3BA^4B\right] = \left[A^2E_3\right], & \tilde{\epsilon}_2 &= \frac{11}{774144},\\
F_3 &= \left[A^2BA^5B\right] = \left[A^2E_4\right], & \tilde{\epsilon}_3 &= -\frac{41}{3225600},\\
F_4 &= \left[ABA^6B\right] = \left[ABE_1\right], & \tilde{\epsilon}_4 &= \frac{17}{9676800},\\
F_5 &= \left[BA^7B\right] = \left[BAE_1\right], & \tilde{\epsilon}_5 &= \frac{1}{302400},\\
F_6 &= \left[A^3B^2A^3B\right] = \left[A^2E_7\right], & \tilde{\epsilon}_6 &= \frac{1}{290304},\\
F_7 &= \left[A^2BABA^3B\right] = \left[A^2E_8\right], & \tilde{\epsilon}_7 &= -\frac{1}{48384},
\end{aligned}
$$

$$
\begin{aligned}
F_8 &= \left[A^2BA^3BAB\right] = \left[A^2BA^2BA^2B\right] = \left[A^2E_9\right], & \tilde{\epsilon}_8 &= \frac{1}{26880},\\
F_9 &= \left[ABA^2BA^3B\right] = \left[ABE_2\right], & \tilde{\epsilon}_9 &= -\frac{1}{13824},\\
F_{10} &= \left[ABABA^4B\right] = \left[ABE_3\right], & \tilde{\epsilon}_{10} &= \frac{1}{15360},\\
F_{11} &= \left[AB^2A^5B\right] = \left[ABE_4\right], & \tilde{\epsilon}_{11} &= -\frac{23}{1612800},\\
F_{12} &= \left[BA^3BA^3B\right] = \left[BAE_2\right], & \tilde{\epsilon}_{12} &= \frac{211}{4838400},\\
F_{13} &= \left[BA^2BA^4B\right] = \left[BAE_3\right], & \tilde{\epsilon}_{13} &= -\frac{37}{1209600},\\
F_{14} &= \left[BABA^5B\right] = \left[BAE_4\right], & \tilde{\epsilon}_{14} &= \frac{7}{691200},\\
F_{15} &= \left[A^2BA^2B^2AB\right] = -\left[A^2E_{12}\right], & \tilde{\epsilon}_{15} &= \frac{47}{1209600},\\
F_{16} &= \left[A^2BAB^2A^2B\right] = \left[A^2BABABAB\right] = -\left[A^2E_{13}\right], & \tilde{\epsilon}_{16} &= -\frac{281}{2419200},\\
F_{17} &= \left[A^2B^2ABA^2B\right] = \left[A^2B^2A^2BAB\right] = -\left[A^2E_{14}\right], & \tilde{\epsilon}_{17} &= \frac{163}{1612800},\\
F_{18} &= \left[ABA^2BABAB\right] = \left[ABA^2B^2A^2B\right] = \left[ABE_5\right], & \tilde{\epsilon}_{18} &= \frac{73}{1612800},\\
F_{19} &= \left[ABABA^2BAB\right] = \left[ABABABA^2B\right] = \left[ABE_6\right], & \tilde{\epsilon}_{19} &= -\frac{43}{2419200},\\
F_{20} &= \left[ABAB^2A^3B\right] = \left[ABE_7\right], & \tilde{\epsilon}_{20} &= -\frac{79}{1209600},\\
F_{21} &= \left[AB^2ABA^3B\right] = \left[ABE_8\right], & \tilde{\epsilon}_{21} &= -\frac{1}{193536},\\
F_{22} &= \left[AB^2A^3BAB\right] = \left[AB^2A^2BA^2B\right] = \left[ABE_9\right], & \tilde{\epsilon}_{22} &= -\frac{13}{3225600},\\
F_{23} &= \left[BA^3BABAB\right] = \left[BA^3B^2A^2B\right] = \left[BAE_5\right], & \tilde{\epsilon}_{23} &= -\frac{1}{25200},\\
F_{24} &= \left[BA^2BA^2BAB\right] = \left[BA^2BABA^2B\right] = \left[BAE_6\right], & \tilde{\epsilon}_{24} &= \frac{1}{11200},\\
F_{25} &= \left[BA^2B^2A^3B\right] = \left[BAE_7\right], & \tilde{\epsilon}_{25} &= -\frac{1}{75600},\\
F_{26} &= \left[BABABA^3B\right] = \left[BAE_8\right], & \tilde{\epsilon}_{26} &= \frac{1}{12600},\\
F_{27} &= \left[BABA^3BAB\right] = \left[BABA^2BA^2B\right] = \left[BAE_9\right], & \tilde{\epsilon}_{27} &= -\frac{1}{25200},\\
F_{28} &= \left[B^2ABA^4B\right] = \left[B^2E_3\right], & \tilde{\epsilon}_{28} &= -\frac{1}{67200},
\end{aligned}
$$

$$F_{29} = \left[A^2BAB^4A\right] = \left[A^2E_{16}\right], \qquad \tilde{\epsilon}_{29} = -\frac{1}{67200},$$

$$F_{30} = \left[ABAB^3ABA\right] = \left[ABAB^2AB^2A\right] = \left[ABE_{10}\right], \qquad \tilde{\epsilon}_{30} = -\frac{31}{403200},$$

$$F_{31} = \left[ABABAB^3A\right] = \left[ABE_{11}\right], \qquad \tilde{\epsilon}_{31} = \frac{11}{201600},$$

$$F_{32} = \left[AB^2A^2B^3A\right] = \left[ABE_{12}\right], \qquad \tilde{\epsilon}_{32} = -\frac{1}{1209600},$$

$$F_{33} = \left[AB^2AB^2ABA\right] = \left[AB^2ABAB^2A\right] = \left[ABE_{13}\right], \qquad \tilde{\epsilon}_{33} = \frac{17}{134400},$$

$$F_{34} = \left[AB^3ABABA\right] = \left[AB^3A^2B^2A\right] = \left[ABE_{14}\right], \qquad \tilde{\epsilon}_{34} = -\frac{1}{19200},$$

$$F_{35} = \left[BA^2B^3ABA\right] = \left[BA^2B^2AB^2A\right] = \left[BAE_{10}\right], \qquad \tilde{\epsilon}_{35} = \frac{31}{604800},$$

$$F_{36} = \left[BA^2BAB^3A\right] = \left[BAE_{11}\right], \qquad \tilde{\epsilon}_{36} = -\frac{1}{60480},$$

$$F_{37} = \left[BABA^2B^3A\right] = \left[BAE_{12}\right], \qquad \tilde{\epsilon}_{37} = -\frac{1}{50400},$$

$$F_{38} = \left[BABAB^2ABA\right] = \left[BABABAB^2A\right] = \left[BAE_{13}\right], \quad \tilde{\epsilon}_{38} = -\frac{1}{5400},$$

$$F_{39} = \left[BAB^2ABABA\right] = \left[BAB^2A^2B^2A\right] = \left[BAE_{14}\right], \qquad \tilde{\epsilon}_{39} = \frac{1}{12600},$$

$$F_{40} = \left[B^2A^2BAB^2A\right] = \left[B^2A^2B^2ABA\right] = -\left[B^2E_5\right], \qquad \tilde{\epsilon}_{40} = \frac{1}{12600},$$

$$F_{41} = \left[B^2ABA^2B^2A\right] = \left[B^2ABABABA\right] = -\left[B^2E_6\right], \quad \tilde{\epsilon}_{41} = -\frac{11}{151200},$$

$$F_{42} = \left[B^2AB^2A^2BA\right] = -\left[B^2E_7\right], \qquad \tilde{\epsilon}_{42} = \frac{1}{100800},$$

$$F_{43} = \left[ABAB^5A\right] = \left[ABE_{15}\right], \qquad \tilde{\epsilon}_{43} = -\frac{29}{1209600},$$

$$F_{44} = \left[AB^2AB^4A\right] = \left[ABE_{16}\right], \qquad \tilde{\epsilon}_{44} = \frac{83}{1209600},$$

$$F_{45} = \left[AB^3AB^3A\right] = \left[ABE_{17}\right], \qquad \tilde{\epsilon}_{45} = -\frac{31}{1209600},$$

$$F_{46} = \left[BA^2B^5A\right] = \left[BAE_{15}\right], \qquad \tilde{\epsilon}_{46} = \frac{1}{50400},$$

$$F_{47} = \left[BABAB^4A\right] = \left[BAE_{16}\right], \qquad \tilde{\epsilon}_{47} = -\frac{1}{10080},$$

$$F_{48} = \left[BAB^2AB^3A\right] = \left[BAE_{17}\right], \qquad \tilde{\epsilon}_{48} = \frac{1}{30240},$$

$$F_{49} = \left[B^2AB^3ABA\right] = \left[B^2AB^2AB^2A\right] = \left[B^2E_{10}\right], \qquad \tilde{\epsilon}_{49} = -\frac{1}{20160},$$

$$
\begin{aligned}
F_{50} &= \left[B^2ABAB^3A\right] = \left[B^2E_{11}\right], & \tilde{\epsilon}_{50} &= \frac{1}{15120},\\
F_{51} &= \left[B^3A^2B^3A\right] = \left[B^2E_{12}\right], & \tilde{\epsilon}_{51} &= -\frac{1}{90720},\\
F_{52} &= \left[AB^7A\right] = \left[ABE_{18}\right], & \tilde{\epsilon}_{52} &= \frac{1}{302400},\\
F_{53} &= \left[BAB^6A\right] = \left[BAE_{18}\right], & \tilde{\epsilon}_{53} &= -\frac{1}{75600},\\
F_{54} &= \left[B^2AB^5A\right] = \left[B^2E_{15}\right], & \tilde{\epsilon}_{54} &= \frac{1}{50400},\\
F_{55} &= \left[B^3AB^4A\right] = \left[B^2E_{16}\right], & \tilde{\epsilon}_{55} &= -\frac{1}{60480},\\
F_{56} &= \left[B^8A\right] = \left[B^2E_{18}\right], & \tilde{\epsilon}_{56} &= -\frac{1}{1209600}.
\end{aligned}
$$

$$
\begin{aligned}
\left[A^4BA^3B\right] = \left[A^2E_2\right] &= 3F_2 - 3F_3 + F_4,\\
\left[A^6BAB\right] = \left[A^5BA^2B\right] = \left[A^4D_2\right] &= 5F_2 - 6F_3 + 2F_4,\\
\left[A^4BABAB\right] = \left[A^4B^2A^2B\right] = \left[A^2E_5\right] &= 4F_6 - 14F_7 + 9F_8 - 16F_9 + 22F_{10}\\
&\quad - 4F_{11} + 9F_{12} - 12F_{13} + 3F_{14},\\
\left[A^3BA^2BAB\right] = \left[A^3BABA^2B\right] = \left[A^2E_6\right] &= F_6 - 4F_7 + 4F_8 - 6F_9 + 7F_{10}\\
&\quad - F_{11} + 3F_{12} - 4F_{13} + F_{14},\\
\left[B^2A^6B\right] = \left[B^2E_1\right] &= F_{12} - 3F_{13} + 3F_{14},\\
\left[A^5B^2AB\right] = -\left[A^4D_4\right] &= 10F_6 - 30F_7 + 15F_8 - 30F_9 + 45F_{10}\\
&\quad - 9F_{11} + 18F_{12} - 24F_{13} + 6F_{14},\\
\left[A^2B^2A^4B\right] = \left[A^2B^2D_1\right] &= 2F_7 - F_8,\\
\left[ABA^3BA^2B\right] = \left[ABA^4BAB\right] = \left[ABA^2D_2\right] &= 2F_9 - F_{10},\\
\left[BA^5BAB\right] = \left[BA^4BA^2B\right] = \left[BA^3D_2\right] &= 2F_{12} - F_{13},\\
\left[A^3B^3A^2B\right] = \left[A^3B^2ABAB\right] = -\left[A^2E_{10}\right] &= 2F_{16} - 2F_{19} + F_{22},\\
\left[A^3BAB^2AB\right] = -\left[A^2E_{11}\right] &= 4F_{15} - 5F_{16} + 3F_{17} - 3F_{18}\\
&\quad + 5F_{19} - 4F_{20} + F_{21},\\
\left[B^2A^2BA^3B\right] = \left[B^2E_2\right] &= -F_{15} + 4F_{16} - 3F_{17} - 3F_{18} + 2F_{19}\\
&\quad + 2F_{20} - F_{22} + 2F_{23} - 3F_{24}\\
&\quad - 2F_{26} + 3F_{27} + F_{28},\\
\left[B^3A^5B\right] = \left[B^2E_4\right] &= -2F_{15} + 8F_{16} - 6F_{17} - 6F_{18} + 4F_{19}\\
&\quad + 4F_{20} - 2F_{22} + 3F_{23} - 3F_{24}\\
&\quad + F_{25} - 6F_{26} + 3F_{27} + 3F_{28},
\end{aligned}
$$

$$
\begin{aligned}
\left[A^4B^3AB\right] = -\left[A^4D_6\right] &= 8F_{15} - 12F_{16} + 6F_{17} - 6F_{18} \\
&\quad + 12F_{19} - 8F_{20} + 2F_{21} - F_{22}, \\
\left[A^2B^3A^3B\right] = \left[A^2B^2D_3\right] &= F_{15} - 3F_{16} + 3F_{17}, \\
\left[ABA^3B^2AB\right] = -\left[ABA^2D_4\right] &= 3F_{18} - 3F_{19} + F_{20}, \\
\left[AB^3A^4B\right] = \left[AB^3D_1\right] &= 2F_{21} - F_{22}, \\
\left[BA^4B^2AB\right] = -\left[BA^3D_4\right] &= 3F_{23} - 3F_{24} + F_{25}, \\
\left[BAB^2A^4B\right] = \left[BAB^2D_1\right] &= 2F_{26} - F_{27}, \\
\left[B^2A^4BAB\right] = \left[B^2A^3BA^2B\right] = \left[B^2A^2D_2\right] &= -2F_{15} + 8F_{16} - 6F_{17} - 6F_{18} + 4F_{19} \\
&\quad + 4F_{20} - 2F_{22} + 4F_{23} - 6F_{24} \\
&\quad - 4F_{26} + 6F_{27} + F_{28}, \\
\left[A^2B^4ABA\right] = \left[A^2B^3AB^2A\right] = \left[A^2B^2D_5\right] &= -2F_{42} + 8F_{41} - 6F_{40} - 6F_{39} + 4F_{38} \\
&\quad + 4F_{37} - 2F_{35} + 4F_{34} - 6F_{33} \\
&\quad - 4F_{31} + 6F_{30} + F_{29}, \\
\left[ABA^2B^4A\right] = \left[ABA^2D_6\right] &= 2F_{31} - F_{30}, \\
\left[AB^4A^2BA\right] = -\left[AB^3D_3\right] &= 3F_{34} - 3F_{33} + F_{32}, \\
\left[BA^3B^4A\right] = \left[BA^3D_6\right] &= 2F_{36} - F_{35}, \\
\left[BAB^3A^2BA\right] = -\left[BAB^2D_3\right] &= 3F_{39} - 3F_{38} + F_{37}, \\
\left[B^2A^3B^3A\right] = \left[B^2A^2D_4\right] &= F_{42} - 3F_{41} + 3F_{40}, \\
\left[B^4A^3BA\right] = -\left[B^4D_1\right] &= 8F_{42} - 12F_{41} + 6F_{40} - 6F_{39} \\
&\quad + 12F_{38} - 8F_{37} + 2F_{36} - F_{35}, \\
\left[A^3B^5A\right] = \left[A^2E_{15}\right] &= -2F_{42} + 8F_{41} - 6F_{40} - 6F_{39} + 4F_{38} \\
&\quad + 4F_{37} - 2F_{35} + 3F_{34} - 3F_{33} \\
&\quad + F_{32} - 6F_{31} + 3F_{30} + 3F_{29}, \\
\left[A^2B^2AB^3A\right] = \left[A^2E_{17}\right] &= -F_{42} + 4F_{41} - 3F_{40} - 3F_{39} + 2F_{38} \\
&\quad + 2F_{37} - F_{35} + 2F_{34} - 3F_{33} \\
&\quad - 2F_{31} + 3F_{30} + F_{29}, \\
\left[B^3ABA^2BA\right] = -\left[B^2E_8\right] &= 4F_{42} - 5F_{41} + 3F_{40} - 3F_{39} \\
&\quad + 5F_{38} - 4F_{37} + F_{36}, \\
\left[B^3A^3B^2A\right] = \left[B^3A^2BABA\right] = -\left[B^2E_9\right] &= 2F_{41} - 2F_{38} + F_{35}, \\
\left[AB^5ABA\right] = \left[AB^4AB^2A\right] = \left[AB^3D_5\right] &= 2F_{45} - F_{44}, \\
\left[BAB^3AB^2A\right] = \left[BAB^4ABA\right] = \left[BAB^2D_5\right] &= 2F_{48} - F_{47}, \\
\left[B^2A^2B^4A\right] = \left[B^2A^2D_6\right] &= 2F_{50} - F_{49},
\end{aligned}
$$

$$
\begin{aligned}
\left[B^5A^2BA\right] = -\left[B^4D_3\right] &= 10F_{51} - 30F_{50} + 15F_{49} - 30F_{48} + 45F_{47} \\
&\quad - 9F_{46} + 18F_{45} - 24F_{44} + 6F_{43}, \\
\left[A^2B^6A\right] = \left[A^2E_{18}\right] &= F_{45} - 3F_{44} + 3F_{43}, \\
\left[B^3AB^2ABA\right] = \left[B^3ABAB^2A\right] = \left[B^2E_{13}\right] &= F_{51} - 4F_{50} + 4F_{49} - 6F_{48} + 7F_{47} \\
&\quad - F_{46} + 3F_{45} - 4F_{44} + F_{43}, \\
\left[B^4ABABA\right] = \left[B^4A^2B^2A\right] = \left[B^2E_{14}\right] &= 4F_{51} - 14F_{50} + 9F_{49} - 16F_{48} + 22F_{47} \\
&\quad - 4F_{46} + 9F_{45} - 12F_{44} + 3F_{43}, \\
\left[B^6ABA\right] = \left[B^5AB^2A\right] = \left[B^4D_5\right] &= 5F_{55} - 6F_{54} + 2F_{53}, \\
\left[B^4AB^3A\right] = \left[B^2E_{17}\right] &= 3F_{55} - 3F_{54} + F_{53}.
\end{aligned}
$$

Coefficient recursive formulae

2nd order

$$
\alpha_A^{(i,i-1)} = \alpha_B^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \sigma^{(i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \sigma^{(i-1)}, \tag{3}
$$

$$
\beta_A^{(i,i-1)} = \beta_B^{(i-1,i-1)} + \tilde{\beta} a_i \left(\sigma^{(i-1)}\right)^2. \tag{4}
$$

4th order

$$
\begin{aligned}
\gamma_{1,A}^{(i,i-1)} &= \gamma_{1,B}^{(i-1,i-1)} + \tilde{\gamma}_1 a_i^4 \sigma^{(i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \sigma^{(i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \\
&\quad - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} + \tilde{\alpha} a_i^2 \alpha_B^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \alpha_B^{(i-1,i-1)},
\end{aligned} \tag{5}
$$

$$
\begin{aligned}
\gamma_{2,A}^{(i,i-1)} &= \gamma_{2,B}^{(i-1,i-1)} + \tilde{\gamma}_2 a_i^3 \left(\sigma^{(i-1)}\right)^2 - \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&\quad - \tilde{\alpha} a_i^2 \beta_B^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
\gamma_{3,A}^{(i,i-1)} &= \gamma_{3,B}^{(i-1,i-1)} + \tilde{\gamma}_3 a_i^3 \left(\sigma^{(i-1)}\right)^2 - \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&\quad - 2\tilde{\beta} a_i \sigma^{(i-1)} \alpha_B^{(i-1,i-1)},
\end{aligned} \tag{7}
$$

$$
\gamma_{4,A}^{(i,i-1)} = \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)}\right)^3 + \tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 + \tilde{\beta} a_i \sigma^{(i-1)} \beta_B^{(i-1,i-1)}, \tag{8}
$$

$$
\gamma_{5,A}^{(i,i-1)} = \gamma_{5,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^3 + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 - 2\tilde{\beta} a_i \sigma^{(i-1)} \beta_B^{(i-1,i-1)}, \tag{9}
$$

$$
\gamma_{6,A}^{(i,i-1)} = \gamma_{6,B}^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^4. \tag{10}
$$

6th order

$$
\begin{aligned}
\delta_{1,A}^{(i,i-1)} &= \delta_{1,B}^{(i-1,i-1)} + \tilde{\delta}_1 a_i^6 \sigma^{(i-1)} + \left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4\right) a_i^5 \nu^{(i-1)} \sigma^{(i-1)} \\
&+ \left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \\
&- \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} - \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \\
&- \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^5 \sigma^{(i-1)} + \tilde{\gamma}_1 a_i^4 \alpha_B^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \gamma_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{11}
$$

$$
\begin{aligned}
\delta_{2,A}^{(i,i-1)} &= \delta_{2,B}^{(i-1,i-1)} + \tilde{\delta}_2 a_i^5 \left(\sigma^{(i-1)}\right)^2 + \left(3\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + 2\tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \\
&- \left(3\tilde{\delta}_{10} + 4\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 3\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 4\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 - 2\tilde{\gamma}_1 a_i^4 \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ 2\tilde{\alpha} a_i^2 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \gamma_{3,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \left(\alpha_B^{(i-1,i-1)}\right)^2,
\end{aligned} \tag{12}
$$

$$
\begin{aligned}
\delta_{3,A}^{(i,i-1)} &= \delta_{3,B}^{(i-1,i-1)} + \tilde{\delta}_3 a_i^5 \left(\sigma^{(i-1)}\right)^2 - \left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \\
&+ \left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 + \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \\
&+ \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 + \tilde{\gamma}_1 a_i^4 \beta_B^{(i-1,i-1)} + \left(\tilde{\gamma}_2 - 2\tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \beta_B^{(i-1,i-1)} - \left(2\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \left(\alpha_B^{(i-1,i-1)}\right)^2,
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
\delta_{4,A}^{(i,i-1)} &= \delta_{4,B}^{(i-1,i-1)} + \tilde{\delta}_4 a_i^5 \left(\sigma^{(i-1)}\right)^2 + \left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \\
&- \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 - \left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \\
&- \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 + 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \left(\alpha_B^{(i-1,i-1)}\right)^2,
\end{aligned} \tag{14}
$$

$$
\begin{aligned}
\delta_{5,A}^{(i,i-1)} &= \delta_{5,B}^{(i-1,i-1)} + \tilde{\delta}_5 a_i^4 \left(\sigma^{(i-1)}\right)^3 - \left(\tilde{\delta}_{10} + 3\tilde{\delta}_{11} + 3\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 + \tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 3\tilde{\alpha} a_i^2 \gamma_{4,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 3\tilde{\beta} a_i \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{15}
$$

$$
\begin{aligned}
\delta_{6,A}^{(i,i-1)} &= \delta_{6,B}^{(i-1,i-1)} + \tilde{\delta}_6 a_i^4 \left(\sigma^{(i-1)}\right)^3 - \left(\tilde{\delta}_{10} - \tilde{\delta}_{11} - 3\tilde{\delta}_{12} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(\tilde{\delta}_{15} + 3\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 + \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 - 3\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(3\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(6\tilde{\gamma}_4 + 3\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 3\tilde{\alpha} a_i^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{16}
$$

$$
\begin{aligned}
\delta_{7,A}^{(i,i-1)} &= \delta_{7,B}^{(i-1,i-1)} + \tilde{\delta}_7 a_i^4 \left(\sigma^{(i-1)}\right)^3 - \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(2\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{17}
$$

$$
\begin{aligned}
\delta_{8,A}^{(i,i-1)} &= \delta_{8,B}^{(i-1,i-1)} + \tilde{\delta}_8 a_i^4 \left(\sigma^{(i-1)}\right)^3 - \left(\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 - \left(\tilde{\delta}_{16} + 2\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- 3\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 - 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{18}
$$

$$
\begin{aligned}
\delta_{9,A}^{(i,i-1)} &= \delta_{9,B}^{(i-1,i-1)} + \tilde{\delta}_9 a_i^4 \left(\sigma^{(i-1)}\right)^3 - \left(2\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 - \left(2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 - 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{19}
$$

$$
\begin{aligned}
\delta_{10,A}^{(i,i-1)} &= \delta_{10,B}^{(i-1,i-1)} + \tilde{\delta}_{10} a_i^3 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_{15} - \tilde{\delta}_{16}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 + \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
&- \left(2\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \tilde{\beta} a_i \left(\beta_B^{(i-1,i-1)}\right)^2,
\end{aligned} \tag{20}
$$

$$\begin{aligned}\delta_{11,A}^{(i,i-1)} &= \delta_{11,B}^{(i-1,i-1)} + \tilde{\delta}_{11}a_i^3\left(\sigma^{(i-1)}\right)^4 + \left(3\tilde{\delta}_{15}+2\tilde{\delta}_{16}+\tilde{\delta}_{17}\right)a_i^2\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^4 + 3\tilde{\delta}_{18}a_i\left(\nu^{(i-1)}\right)^2\left(\sigma^{(i-1)}\right)^4 \\ &+ \tilde{\gamma}_4 a_i^2\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} + \tilde{\gamma}_6 a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} + 2\tilde{\alpha}a_i^2\gamma_{6,B}^{(i-1,i-1)} \\ &+ \tilde{\beta}a_i\sigma^{(i-1)}\gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\beta}a_i\nu^{(i-1)}\gamma_{6,B}^{(i-1,i-1)},\end{aligned} \tag{21}$$

$$\begin{aligned}\delta_{12,A}^{(i,i-1)} &= \delta_{12,B}^{(i-1,i-1)} + \tilde{\delta}_{12}a_i^3\left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_{16}+\tilde{\delta}_{17}\right)a_i^2\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^4 + 2\tilde{\delta}_{18}a_i\left(\nu^{(i-1)}\right)^2\left(\sigma^{(i-1)}\right)^4 \\ &+ \tilde{\gamma}_5 a_i^2\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i\left(\sigma^{(i-1)}\right)^3\alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} \\ &- 2\tilde{\beta}a_i\sigma^{(i-1)}\gamma_{4,B}^{(i-1,i-1)},\end{aligned} \tag{22}$$

$$\begin{aligned}\delta_{13,A}^{(i,i-1)} &= \delta_{13,B}^{(i-1,i-1)} + \tilde{\delta}_{13}a_i^3\left(\sigma^{(i-1)}\right)^4 + 2\tilde{\delta}_{16}a_i^2\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^4 - \tilde{\delta}_{18}a_i\left(\nu^{(i-1)}\right)^2\left(\sigma^{(i-1)}\right)^4 \\ &- 4\tilde{\gamma}_5 a_i^2\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i\left(\sigma^{(i-1)}\right)^3\alpha_B^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} \\ &- 2\tilde{\beta}a_i\sigma^{(i-1)}\gamma_{5,B}^{(i-1,i-1)} + 2\tilde{\beta}a_i\left(\beta_B^{(i-1,i-1)}\right)^2,\end{aligned} \tag{23}$$

$$\begin{aligned}\delta_{14,A}^{(i,i-1)} &= \delta_{14,B}^{(i-1,i-1)} + \tilde{\delta}_{14}a_i^3\left(\sigma^{(i-1)}\right)^4 + 2\tilde{\delta}_{17}a_i^2\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^4 + 5\tilde{\delta}_{18}a_i\left(\nu^{(i-1)}\right)^2\left(\sigma^{(i-1)}\right)^4 \\ &+ \tilde{\gamma}_5 a_i^2\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i\left(\sigma^{(i-1)}\right)^3\alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^2\beta_B^{(i-1,i-1)} \\ &- \tilde{\beta}a_i\left(\beta_B^{(i-1,i-1)}\right)^2,\end{aligned} \tag{24}$$

$$\begin{aligned}\delta_{15,A}^{(i,i-1)} &= \delta_{15,B}^{(i-1,i-1)} + \tilde{\delta}_{15}a_i^2\left(\sigma^{(i-1)}\right)^5 + \tilde{\delta}_{18}a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^5 + \tilde{\gamma}_6 a_i\left(\sigma^{(i-1)}\right)^3\beta_B^{(i-1,i-1)} \\ &+ \tilde{\beta}a_i\sigma^{(i-1)}\gamma_{6,B}^{(i-1,i-1)},\end{aligned} \tag{25}$$

$$\delta_{16,A}^{(i,i-1)} = \delta_{16,B}^{(i-1,i-1)} + \tilde{\delta}_{16}a_i^2\left(\sigma^{(i-1)}\right)^5 - \tilde{\delta}_{18}a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^5 - 2\tilde{\beta}a_i\sigma^{(i-1)}\gamma_{6,B}^{(i-1,i-1)}, \tag{26}$$

$$\delta_{17,A}^{(i,i-1)} = \delta_{17,B}^{(i-1,i-1)} + \tilde{\delta}_{17}a_i^2\left(\sigma^{(i-1)}\right)^5 + 5\tilde{\delta}_{18}a_i\nu^{(i-1)}\left(\sigma^{(i-1)}\right)^5 - 2\tilde{\gamma}_6 a_i\left(\sigma^{(i-1)}\right)^3\beta_B^{(i-1,i-1)}, \tag{27}$$

$$\delta_{18,A}^{(i,i-1)} = \delta_{18,B}^{(i-1,i-1)} + \tilde{\delta}_{18}a_i\left(\sigma^{(i-1)}\right)^6. \tag{28}$$

8th order

$$
\begin{aligned}
\epsilon_{1,A}^{(i,i-1)} &= \epsilon_{1,B}^{(i-1,i-1)} + \tilde{\epsilon}_1 a_i^8 \sigma^{(i-1)} + \left(\tilde{\epsilon}_2 + \tilde{\epsilon}_3 + \tilde{\epsilon}_4 + \tilde{\epsilon}_5\right) a_i^7 \nu^{(i-1)} \sigma^{(i-1)} \\
&+ \left(\tilde{\epsilon}_6 + \tilde{\epsilon}_7 + \tilde{\epsilon}_8 + \tilde{\epsilon}_9 + \tilde{\epsilon}_{10} + \tilde{\epsilon}_{11} + \tilde{\epsilon}_{12} + \tilde{\epsilon}_{13} + \tilde{\epsilon}_{14}\right) a_i^6 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \\
&+ \left(\tilde{\epsilon}_{15} + \tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + \tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} + \tilde{\epsilon}_{20} + \tilde{\epsilon}_{21} + \tilde{\epsilon}_{22} + \tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + \tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}\right) a_i^5 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \\
&- \left(\tilde{\epsilon}_{29} + \tilde{\epsilon}_{30} + \tilde{\epsilon}_{31} + \tilde{\epsilon}_{32} + \tilde{\epsilon}_{33} + \tilde{\epsilon}_{34} + \tilde{\epsilon}_{35} + \tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + \tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \\
&- \left(\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} + \tilde{\epsilon}_{50} + \tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^5 \sigma^{(i-1)} - \left(\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^6 \sigma^{(i-1)} \\
&- \tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^7 \sigma^{(i-1)} + \tilde{\delta}_1 a_i^6 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4\right) a_i^5 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} + \tilde{\gamma}_1 a_i^4 \gamma_{1,B}^{(i-1,i-1)} \\
&+ \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&+ \tilde{\alpha} a_i^2 \delta_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{1,B}^{(i-1,i-1)},
\end{aligned}
\tag{29}
$$

$$
\begin{aligned}
\epsilon_{2,A}^{(i,i-1)} &= \epsilon_{2,B}^{(i-1,i-1)} + \tilde{\epsilon}_2 a_i^7 \left(\sigma^{(i-1)}\right)^2 + \left(4\tilde{\epsilon}_6 + 3\tilde{\epsilon}_7 + 5\tilde{\epsilon}_8 + 3\tilde{\epsilon}_9 + \tilde{\epsilon}_{10} + 3\tilde{\epsilon}_{12} + \tilde{\epsilon}_{13}\right) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \\
&+ \left(10\tilde{\epsilon}_{15} + 8\tilde{\epsilon}_{16} + 6\tilde{\epsilon}_{17} + 8\tilde{\epsilon}_{18} + 6\tilde{\epsilon}_{19} + 4\tilde{\epsilon}_{20} + 3\tilde{\epsilon}_{21} + 5\tilde{\epsilon}_{22} + 8\tilde{\epsilon}_{23} + 6\tilde{\epsilon}_{24} + 4\tilde{\epsilon}_{25} + 3\tilde{\epsilon}_{26} + 5\tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}\right) a_i^5 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(13\tilde{\epsilon}_{29} + 9\tilde{\epsilon}_{30} + 11\tilde{\epsilon}_{31} + 10\tilde{\epsilon}_{32} + 8\tilde{\epsilon}_{33} + 6\tilde{\epsilon}_{34} + 9\tilde{\epsilon}_{35} + 11\tilde{\epsilon}_{36} + 10\tilde{\epsilon}_{37} + 8\tilde{\epsilon}_{38} + 6\tilde{\epsilon}_{39} + 8\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 4\tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(14\tilde{\epsilon}_{43} + 13\tilde{\epsilon}_{44} + 11\tilde{\epsilon}_{45} + 14\tilde{\epsilon}_{46} + 13\tilde{\epsilon}_{47} + 11\tilde{\epsilon}_{48} + 9\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 10\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} + 14\tilde{\epsilon}_{54} + 13\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^2 - 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^6 \left(\sigma^{(i-1)}\right)^2 - 5\tilde{\delta}_1 a_i^6 \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_2 + \tilde{\delta}_3\right) a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 5\left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4\right) a_i^5 \nu^{(i-1)} \beta_B^{(i-1,i-1)} + \left(2\tilde{\delta}_5 + 2\tilde{\delta}_6 + 2\tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 5\left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{10} + 3\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + 2\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 5\left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - \left(4\tilde{\delta}_{15} + 3\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 5\left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^5 \beta_B^{(i-1,i-1)} \\
&+ 5\tilde{\gamma}_1 a_i^4 \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_1 a_i^4 \gamma_{3,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 5\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&+ 3\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 5\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- 3\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} + \tilde{\gamma}_2 a_i^3 \left(\alpha_B^{(i-1,i-1)}\right)^2 - \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ 3\tilde{\alpha} a_i^2 \delta_{2,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{3,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{2,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{3,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned}
\tag{30}
$$

$$
\begin{aligned}
\epsilon_{3,A}^{(i,i-1)} &= \epsilon_{3,B}^{(i-1,i-1)} + \tilde{\epsilon}_3 a_i^7 \left(\sigma^{(i-1)}\right)^2 - \left(3\tilde{\epsilon}_6 + 2\tilde{\epsilon}_7 + 5\tilde{\epsilon}_8 + 3\tilde{\epsilon}_9 - \tilde{\epsilon}_{11} + 3\tilde{\epsilon}_{12} - \tilde{\epsilon}_{14}\right) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \\
&- \left(11\tilde{\epsilon}_{15} + 8\tilde{\epsilon}_{16} + 5\tilde{\epsilon}_{17} + 9\tilde{\epsilon}_{18} + 6\tilde{\epsilon}_{19} + 3\tilde{\epsilon}_{20} + 2\tilde{\epsilon}_{21} + 5\tilde{\epsilon}_{22} + 9\tilde{\epsilon}_{23} + 6\tilde{\epsilon}_{24} + 3\tilde{\epsilon}_{25} + 2\tilde{\epsilon}_{26} + 5\tilde{\epsilon}_{27}\right) a_i^5 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&+ \left(14\tilde{\epsilon}_{29} + 9\tilde{\epsilon}_{30} + 12\tilde{\epsilon}_{31} + 11\tilde{\epsilon}_{32} + 8\tilde{\epsilon}_{33} + 5\tilde{\epsilon}_{34} + 9\tilde{\epsilon}_{35} + 12\tilde{\epsilon}_{36} + 11\tilde{\epsilon}_{37} + 8\tilde{\epsilon}_{38} + 5\tilde{\epsilon}_{39} + 9\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \\
&+ \left(15\tilde{\epsilon}_{43} + 14\tilde{\epsilon}_{44} + 11\tilde{\epsilon}_{45} + 15\tilde{\epsilon}_{46} + 14\tilde{\epsilon}_{47} + 11\tilde{\epsilon}_{48} + 9\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 11\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 \\
&+ \left(14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} + 15\tilde{\epsilon}_{54} + 14\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^2 + 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^6 \left(\sigma^{(i-1)}\right)^2 + 6\tilde{\delta}_1 a_i^6 \beta_B^{(i-1,i-1)} \\
&- \left(\tilde{\delta}_2 + 2\tilde{\delta}_3 - \tilde{\delta}_4\right) a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 6\left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4\right) a_i^5 \nu^{(i-1)} \beta_B^{(i-1,i-1)} - \left(\tilde{\delta}_5 + 2\tilde{\delta}_6 + 3\tilde{\delta}_7 - \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 6\left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(3\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - \tilde{\delta}_{12} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 6\left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + \left(3\tilde{\delta}_{15} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 6\left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 6\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^5 \beta_B^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_1 a_i^4 \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_1 a_i^4 \gamma_{3,B}^{(i-1,i-1)} + \left(2\tilde{\gamma}_2 + 3\tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 6\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&- 3\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - \left(4\tilde{\gamma}_4 + 5\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&+ 3\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} \\
&+ 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} - \left(2\tilde{\gamma}_2 - \tilde{\gamma}_3\right) a_i^3 \left(\alpha_B^{(i-1,i-1)}\right)^2 + \left(4\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- 3\tilde{\alpha} a_i^2 \delta_{2,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{4,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{2,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{4,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned}
\tag{31}
$$

$$
\begin{aligned}
\epsilon_{4,A}^{(i,i-1)} &= \epsilon_{4,B}^{(i-1,i-1)} + \tilde{\epsilon}_4 a_i^7 \left(\sigma^{(i-1)}\right)^2 + \left(\tilde{\epsilon}_6 + \tilde{\epsilon}_7 + 2\tilde{\epsilon}_8 + 2\tilde{\epsilon}_9 + \tilde{\epsilon}_{10} + \tilde{\epsilon}_{11} + \tilde{\epsilon}_{12}\right) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \\
&+ \left(4\tilde{\epsilon}_{15} + 3\tilde{\epsilon}_{16} + 2\tilde{\epsilon}_{17} + 4\tilde{\epsilon}_{18} + 3\tilde{\epsilon}_{19} + 2\tilde{\epsilon}_{20} + 2\tilde{\epsilon}_{21} + 3\tilde{\epsilon}_{22} + 3\tilde{\epsilon}_{23} + 2\tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + 2\tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}\right) a_i^5 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(5\tilde{\epsilon}_{29} + 4\tilde{\epsilon}_{30} + 5\tilde{\epsilon}_{31} + 5\tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{33} + 3\tilde{\epsilon}_{34} + 3\tilde{\epsilon}_{35} + 4\tilde{\epsilon}_{36} + 4\tilde{\epsilon}_{37} + 3\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} + 4\tilde{\epsilon}_{40} + 3\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(6\tilde{\epsilon}_{43} + 6\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 5\tilde{\epsilon}_{46} + 5\tilde{\epsilon}_{47} + 4\tilde{\epsilon}_{48} + 4\tilde{\epsilon}_{49} + 5\tilde{\epsilon}_{50} + 5\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(6\tilde{\epsilon}_{52} + 5\tilde{\epsilon}_{53} + 6\tilde{\epsilon}_{54} + 6\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^2 - 6\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^6 \left(\sigma^{(i-1)}\right)^2 - 2\tilde{\delta}_1 a_i^6 \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_2 + 2\tilde{\delta}_3 - 2\tilde{\delta}_4\right) a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4\right) a_i^5 \nu^{(i-1)} \beta_B^{(i-1,i-1)} + \left(\tilde{\delta}_5 + 2\tilde{\delta}_6 + 3\tilde{\delta}_7 - \tilde{\delta}_8 - 2\tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 2\left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - 2\tilde{\delta}_{12} - \tilde{\delta}_{13}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 2\left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{15} - \tilde{\delta}_{16}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 2\left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^5 \beta_B^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_1 a_i^4 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\gamma}_1 a_i^4 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 2\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&+ 3\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
&+ 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} + \tilde{\gamma}_2 a_i^3 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- 2\tilde{\gamma}_3 a_i^3 \left(\alpha_B^{(i-1,i-1)}\right)^2 - \left(2\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + \tilde{\alpha} a_i^2 \delta_{2,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{1,B}^{(i-1,i-1)} \\
&- \tilde{\beta} a_i \nu^{(i-1)} \delta_{2,B}^{(i-1,i-1)} + 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned}
\tag{32}
$$

$$
\begin{aligned}
\epsilon_{5,A}^{(i,i-1)} &= \epsilon_{5,B}^{(i-1,i-1)} + \tilde{\epsilon}_5 a_i^7 \left(\sigma^{(i-1)}\right)^2 + \left(\tilde{\epsilon}_{12} + \tilde{\epsilon}_{13} + \tilde{\epsilon}_{14}\right) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 + \left(\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + \tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}\right) a_i^5 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(\tilde{\epsilon}_{35} + \tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + \tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 - \left(\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} + \tilde{\epsilon}_{50} + \tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^2 \\
&- \left(\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^2 - \tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^6 \left(\sigma^{(i-1)}\right)^2 + 2\tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 2\left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 2\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + \tilde{\gamma}_3 a_i^3 \left(\alpha_B^{(i-1,i-1)}\right)^2 - \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{1,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned}
\tag{33}
$$

$$
\begin{aligned}
\epsilon_{6,A}^{(i,i-1)} &= \epsilon_{6,B}^{(i-1,i-1)} + \tilde{\epsilon}_6 a_i^6 \left(\sigma^{(i-1)}\right)^3 + \left(10\tilde{\epsilon}_{15} + 4\tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + 4\tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} + \tilde{\epsilon}_{20} + 4\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{25}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(18\tilde{\epsilon}_{29} + 6\tilde{\epsilon}_{30} + 12\tilde{\epsilon}_{31} + 10\tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{33} + \tilde{\epsilon}_{34} + 6\tilde{\epsilon}_{35} + 12\tilde{\epsilon}_{36} + 10\tilde{\epsilon}_{37} + 4\tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + 4\tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(21\tilde{\epsilon}_{43} + 18\tilde{\epsilon}_{44} + 12\tilde{\epsilon}_{45} + 21\tilde{\epsilon}_{46} + 18\tilde{\epsilon}_{47} + 12\tilde{\epsilon}_{48} + 6\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 10\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(21\tilde{\epsilon}_{52} + 21\tilde{\epsilon}_{53} + 21\tilde{\epsilon}_{54} + 18\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 - 21\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 + \tilde{\delta}_2 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(2\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(3\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + 2\tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(4\tilde{\delta}_{10} + 6\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 2\tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{10} + 4\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 3\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(11\tilde{\delta}_{15} + 8\tilde{\delta}_{16} + 6\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 4\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 11\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 10\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 10\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 4\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&+ 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 10\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&+ 10\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - \tilde{\gamma}_2 a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ 4\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{7,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} \\
&- \tilde{\beta} a_i \nu^{(i-1)} \delta_{7,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned}
\tag{34}
$$

$$
\begin{aligned}
\epsilon_{7,A}^{(i,i-1)} &= \epsilon_{7,B}^{(i-1,i-1)} + \tilde{\epsilon}_7 a_i^6 \left(\sigma^{(i-1)}\right)^3 - \left(30\tilde{\epsilon}_{15} + 13\tilde{\epsilon}_{16} + 2\tilde{\epsilon}_{17} + 14\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} - \tilde{\epsilon}_{21} + 14\tilde{\epsilon}_{23} + 4\tilde{\epsilon}_{24} - \tilde{\epsilon}_{26}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(57\tilde{\epsilon}_{29} + 18\tilde{\epsilon}_{30} + 38\tilde{\epsilon}_{31} + 30\tilde{\epsilon}_{32} + 13\tilde{\epsilon}_{33} + 2\tilde{\epsilon}_{34} + 18\tilde{\epsilon}_{35} + 38\tilde{\epsilon}_{36} + 30\tilde{\epsilon}_{37} + 13\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} + 14\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(66\tilde{\epsilon}_{43} + 57\tilde{\epsilon}_{44} + 36\tilde{\epsilon}_{45} + 66\tilde{\epsilon}_{46} + 57\tilde{\epsilon}_{47} + 36\tilde{\epsilon}_{48} + 18\tilde{\epsilon}_{49} + 38\tilde{\epsilon}_{50} + 30\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(63\tilde{\epsilon}_{52} + 63\tilde{\epsilon}_{53} + 66\tilde{\epsilon}_{54} + 57\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 + 63\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 + 2\tilde{\delta}_3 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(4\tilde{\delta}_5 + 4\tilde{\delta}_6 + 2\tilde{\delta}_7 - \tilde{\delta}_8\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(10\tilde{\delta}_{10} + 10\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + 3\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(20\tilde{\delta}_{15} + 9\tilde{\delta}_{16} + 8\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ 17\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 30\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 14\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 4\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + \left(2\tilde{\gamma}_2 + 3\tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 30\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 14\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&- 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} - 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - \left(4\tilde{\gamma}_4 + 5\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&- 30\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 14\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
&- 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 30\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} - 14\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 6\tilde{\gamma}_2 a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6\left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - 14\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - 4\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{8,B}^{(i-1,i-1)} \\
&+ 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} + 14\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} \\
&+ 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{8,B}^{(i-1,i-1)},
\end{aligned}
\tag{35}
$$

$$
\begin{aligned}
\epsilon_{8,A}^{(i,i-1)} &= \epsilon_{8,B}^{(i-1,i-1)} + \tilde{\epsilon}_8 a_i^6 \left(\sigma^{(i-1)}\right)^3 + \left(17\tilde{\epsilon}_{15} + 10\tilde{\epsilon}_{16} + 4\tilde{\epsilon}_{17} + 9\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} + \tilde{\epsilon}_{22} + 9\tilde{\epsilon}_{23} + 4\tilde{\epsilon}_{24} + \tilde{\epsilon}_{27}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(35\tilde{\epsilon}_{29} + 13\tilde{\epsilon}_{30} + 23\tilde{\epsilon}_{31} + 17\tilde{\epsilon}_{32} + 10\tilde{\epsilon}_{33} + 4\tilde{\epsilon}_{34} + 13\tilde{\epsilon}_{35} + 23\tilde{\epsilon}_{36} + 17\tilde{\epsilon}_{37} + 10\tilde{\epsilon}_{38} + 4\tilde{\epsilon}_{39} + 9\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(41\tilde{\epsilon}_{43} + 35\tilde{\epsilon}_{44} + 24\tilde{\epsilon}_{45} + 41\tilde{\epsilon}_{46} + 35\tilde{\epsilon}_{47} + 24\tilde{\epsilon}_{48} + 13\tilde{\epsilon}_{49} + 23\tilde{\epsilon}_{50} + 17\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(42\tilde{\epsilon}_{52} + 42\tilde{\epsilon}_{53} + 41\tilde{\epsilon}_{54} + 35\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 - 42\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 - \left(2\tilde{\delta}_2 + \tilde{\delta}_3\right) a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(5\tilde{\delta}_5 + 4\tilde{\delta}_6 + 3\tilde{\delta}_7 + 2\tilde{\delta}_8 + 3\tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(3\tilde{\delta}_{10} + \tilde{\delta}_{11} - \tilde{\delta}_{12} + 2\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(6\tilde{\delta}_{10} + 7\tilde{\delta}_{11} + 6\tilde{\delta}_{12} + 5\tilde{\delta}_{13} + 4\tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(3\tilde{\delta}_{15} + \tilde{\delta}_{16} + 2\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(9\tilde{\delta}_{15} + 8\tilde{\delta}_{16} + 7\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 9\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 15\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 9\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(2\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 15\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 9\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&+ \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} + \left(4\tilde{\gamma}_4 + 3\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 15\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 9\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} + 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&+ 15\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - \left(3\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ \left(6\tilde{\gamma}_4 + 4\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ 9\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + 4\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{9,B}^{(i-1,i-1)} - 9\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} \\
&- \tilde{\beta} a_i \nu^{(i-1)} \delta_{9,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned}
\tag{36}
$$

$$
\begin{aligned}
\epsilon_{9,A}^{(i,i-1)} &= \epsilon_{9,B}^{(i-1,i-1)} + \tilde{\epsilon}_9 a_i^6 \left(\sigma^{(i-1)}\right)^3 - \left(30\tilde{\epsilon}_{15} + 16\tilde{\epsilon}_{16} + 6\tilde{\epsilon}_{17} + 13\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} - \tilde{\epsilon}_{20} - \tilde{\epsilon}_{21} - 2\tilde{\epsilon}_{22} + 16\tilde{\epsilon}_{23} + 6\tilde{\epsilon}_{24}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(62\tilde{\epsilon}_{29} + 19\tilde{\epsilon}_{30} + 38\tilde{\epsilon}_{31} + 26\tilde{\epsilon}_{32} + 13\tilde{\epsilon}_{33} + 4\tilde{\epsilon}_{34} + 22\tilde{\epsilon}_{35} + 42\tilde{\epsilon}_{36} + 30\tilde{\epsilon}_{37} + 16\tilde{\epsilon}_{38} + 6\tilde{\epsilon}_{39} + 13\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41} - \tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(69\tilde{\epsilon}_{43} + 57\tilde{\epsilon}_{44} + 38\tilde{\epsilon}_{45} + 74\tilde{\epsilon}_{46} + 62\tilde{\epsilon}_{47} + 42\tilde{\epsilon}_{48} + 19\tilde{\epsilon}_{49} + 38\tilde{\epsilon}_{50} + 26\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(69\tilde{\epsilon}_{52} + 74\tilde{\epsilon}_{53} + 69\tilde{\epsilon}_{54} + 57\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 + 69\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 - 2\tilde{\delta}_3 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(4\tilde{\delta}_5 + 4\tilde{\delta}_6 - 2\tilde{\delta}_7 + 2\tilde{\delta}_8 + 4\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(6\tilde{\delta}_{10} + 6\tilde{\delta}_{11} + 6\tilde{\delta}_{12} + 10\tilde{\delta}_{13} + 8\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(12\tilde{\delta}_{15} + 16\tilde{\delta}_{16} + 14\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ 22\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 30\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 16\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 6\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 30\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 16\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&- 6\left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 30\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&- 16\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&- 30\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} - 16\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} + 4\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 7\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 16\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - 6\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{2,B}^{(i-1,i-1)} + 16\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} \\
&+ 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{37}
$$

$$
\begin{aligned}
\epsilon_{10,A}^{(i,i-1)} &= \epsilon_{10,B}^{(i-1,i-1)} + \tilde{\epsilon}_{10} a_i^6 \left(\sigma^{(i-1)}\right)^3 + \left(45\tilde{\epsilon}_{15} + 22\tilde{\epsilon}_{16} + 7\tilde{\epsilon}_{17} + 21\tilde{\epsilon}_{18} + 7\tilde{\epsilon}_{19} + \tilde{\epsilon}_{20} - \tilde{\epsilon}_{22} + 22\tilde{\epsilon}_{23} + 7\tilde{\epsilon}_{24} + \tilde{\epsilon}_{28}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(89\tilde{\epsilon}_{29} + 29\tilde{\epsilon}_{30} + 58\tilde{\epsilon}_{31} + 43\tilde{\epsilon}_{32} + 21\tilde{\epsilon}_{33} + 7\tilde{\epsilon}_{34} + 29\tilde{\epsilon}_{35} + 59\tilde{\epsilon}_{36} + 45\tilde{\epsilon}_{37} + 22\tilde{\epsilon}_{38} + 7\tilde{\epsilon}_{39} + 21\tilde{\epsilon}_{40} + 7\tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(102\tilde{\epsilon}_{43} + 87\tilde{\epsilon}_{44} + 58\tilde{\epsilon}_{45} + 103\tilde{\epsilon}_{46} + 89\tilde{\epsilon}_{47} + 59\tilde{\epsilon}_{48} + 29\tilde{\epsilon}_{49} + 58\tilde{\epsilon}_{50} + 43\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(102\tilde{\epsilon}_{52} + 103\tilde{\epsilon}_{53} + 102\tilde{\epsilon}_{54} + 87\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 - 102\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 - \tilde{\delta}_3 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(8\tilde{\delta}_5 + 7\tilde{\delta}_6 - 2\tilde{\delta}_7 + 2\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(13\tilde{\delta}_{10} + 15\tilde{\delta}_{11} + 5\tilde{\delta}_{12} + 10\tilde{\delta}_{13} + 7\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(29\tilde{\delta}_{15} + 21\tilde{\delta}_{16} + 17\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 31\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 45\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 22\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(7\tilde{\gamma}_2 + 4\tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 45\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 22\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&+ \left(3\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} + \left(14\tilde{\gamma}_4 + 11\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 45\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 22\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} + 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&+ 45\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} + 22\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - 2\left(4\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ 2\left(4\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + \left(16\tilde{\gamma}_4 + 10\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 22\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + 7\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{3,B}^{(i-1,i-1)} - 22\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} \\
&- 7\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned} \tag{38}
$$

$$
\begin{aligned}
\epsilon_{11,A}^{(i,i-1)} &= \epsilon_{11,B}^{(i-1,i-1)} + \tilde{\epsilon}_{11} a_i^6 \left(\sigma^{(i-1)}\right)^3 - \left(9\tilde{\epsilon}_{15} + 4\tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + 4\tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} - \tilde{\epsilon}_{21} - \tilde{\epsilon}_{22} + 4\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(17\tilde{\epsilon}_{29} + 5\tilde{\epsilon}_{30} + 11\tilde{\epsilon}_{31} + 8\tilde{\epsilon}_{32} + 3\tilde{\epsilon}_{33} + 5\tilde{\epsilon}_{35} + 11\tilde{\epsilon}_{36} + 9\tilde{\epsilon}_{37} + 4\tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + 4\tilde{\epsilon}_{40} + \tilde{\epsilon}_{41}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(19\tilde{\epsilon}_{43} + 16\tilde{\epsilon}_{44} + 10\tilde{\epsilon}_{45} + 19\tilde{\epsilon}_{46} + 17\tilde{\epsilon}_{47} + 11\tilde{\epsilon}_{48} + 5\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 8\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(18\tilde{\epsilon}_{52} + 19\tilde{\epsilon}_{53} + 19\tilde{\epsilon}_{54} + 16\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 + 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 + \left(\tilde{\delta}_3 + \tilde{\delta}_4\right) a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(2\tilde{\delta}_5 + \tilde{\delta}_6 - 2\tilde{\delta}_7\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_5 - \tilde{\delta}_7 - \tilde{\delta}_8\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_{10} + 3\tilde{\delta}_{11} + \tilde{\delta}_{12} + 2\tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{11} + \tilde{\delta}_{12} - \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{15} + \tilde{\delta}_{16}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 9\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 4\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&+ 9\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 4\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
&- 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 9\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&+ \left(2\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - \left(4\tilde{\gamma}_4 + 3\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 4\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{4,B}^{(i-1,i-1)} \\
&+ 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned}
\tag{39}
$$

$$
\begin{aligned}
\epsilon_{12,A}^{(i,i-1)} &= \epsilon_{12,B}^{(i-1,i-1)} + \tilde{\epsilon}_{12} a_i^6 \left(\sigma^{(i-1)}\right)^3 + \left(18\tilde{\epsilon}_{15} + 9\tilde{\epsilon}_{16} + 3\tilde{\epsilon}_{17} + 9\tilde{\epsilon}_{18} + 3\tilde{\epsilon}_{19} + 12\tilde{\epsilon}_{23} + 5\tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + 2\tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(36\tilde{\epsilon}_{29} + 12\tilde{\epsilon}_{30} + 24\tilde{\epsilon}_{31} + 18\tilde{\epsilon}_{32} + 9\tilde{\epsilon}_{33} + 3\tilde{\epsilon}_{34} + 15\tilde{\epsilon}_{35} + 28\tilde{\epsilon}_{36} + 22\tilde{\epsilon}_{37} + 12\tilde{\epsilon}_{38} + 5\tilde{\epsilon}_{39} + 13\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(42\tilde{\epsilon}_{43} + 36\tilde{\epsilon}_{44} + 24\tilde{\epsilon}_{45} + 47\tilde{\epsilon}_{46} + 41\tilde{\epsilon}_{47} + 28\tilde{\epsilon}_{48} + 16\tilde{\epsilon}_{49} + 29\tilde{\epsilon}_{50} + 23\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(42\tilde{\epsilon}_{52} + 47\tilde{\epsilon}_{53} + 48\tilde{\epsilon}_{54} + 42\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 - 48\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 - \tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(3\tilde{\delta}_5 + 3\tilde{\delta}_6 + 2\tilde{\delta}_8 + 2\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(6\tilde{\delta}_{10} + 6\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 7\tilde{\delta}_{13} + 7\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 12\left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 20\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 18\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 9\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(3\tilde{\gamma}_2 - \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 18\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 9\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&+ \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} + 2\left(3\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&+ 18\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 9\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
&+ 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(3\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \left(3\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + \left(6\tilde{\gamma}_4 + 4\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 9\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + 3\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} \\
&- 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{2,B}^{(i-1,i-1)} - 9\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} \\
&- \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned}
\tag{40}
$$

$$
\begin{aligned}
\epsilon_{13,A}^{(i,i-1)} &= \epsilon_{13,B}^{(i-1,i-1)} + \tilde{\epsilon}_{13} a_i^6 \left(\sigma^{(i-1)}\right)^3 - \left(24\tilde{\epsilon}_{15} + 12\tilde{\epsilon}_{16} + 4\tilde{\epsilon}_{17} + 12\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} + 13\tilde{\epsilon}_{23} + 4\tilde{\epsilon}_{24} - \tilde{\epsilon}_{25} + \tilde{\epsilon}_{27} + 2\tilde{\epsilon}_{28}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(48\tilde{\epsilon}_{29} + 16\tilde{\epsilon}_{30} + 32\tilde{\epsilon}_{31} + 24\tilde{\epsilon}_{32} + 12\tilde{\epsilon}_{33} + 4\tilde{\epsilon}_{34} + 16\tilde{\epsilon}_{35} + 33\tilde{\epsilon}_{36} + 26\tilde{\epsilon}_{37} + 13\tilde{\epsilon}_{38} + 4\tilde{\epsilon}_{39} + 16\tilde{\epsilon}_{40} + 7\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(56\tilde{\epsilon}_{43} + 48\tilde{\epsilon}_{44} + 32\tilde{\epsilon}_{45} + 57\tilde{\epsilon}_{46} + 50\tilde{\epsilon}_{47} + 33\tilde{\epsilon}_{48} + 19\tilde{\epsilon}_{49} + 36\tilde{\epsilon}_{50} + 29\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&+ \left(56\tilde{\epsilon}_{52} + 57\tilde{\epsilon}_{53} + 60\tilde{\epsilon}_{54} + 53\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 + 60\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 - 2\tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 2\left(2\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_8\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(8\tilde{\delta}_{10} + 8\tilde{\delta}_{11} + 6\tilde{\delta}_{13} + 8\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(16\tilde{\delta}_{15} + 10\tilde{\delta}_{16} + 12\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ 22\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 24\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 12\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 2\left(2\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 24\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 12\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
&- \left(8\tilde{\gamma}_4 + 6\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 24\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 12\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 2\left(2\tilde{\gamma}_2 - \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \left(4\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - 2\left(4\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 12\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - 4\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} \\
&- 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{3,B}^{(i-1,i-1)} + 12\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned}
\tag{41}
$$

$$
\begin{aligned}
\epsilon_{14,A}^{(i,i-1)} &= \epsilon_{14,B}^{(i-1,i-1)} + \tilde{\epsilon}_{14} a_i^6 \left(\sigma^{(i-1)}\right)^3 + \left(6\tilde{\epsilon}_{15} + 3\tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + 3\tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} + 3\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{26} + \tilde{\epsilon}_{27} + 3\tilde{\epsilon}_{28}\right) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \\
&- \left(12\tilde{\epsilon}_{29} + 4\tilde{\epsilon}_{30} + 8\tilde{\epsilon}_{31} + 6\tilde{\epsilon}_{32} + 3\tilde{\epsilon}_{33} + \tilde{\epsilon}_{34} + 4\tilde{\epsilon}_{35} + 8\tilde{\epsilon}_{36} + 7\tilde{\epsilon}_{37} + 4\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} + 6\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}\right) a_i^4 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(14\tilde{\epsilon}_{43} + 12\tilde{\epsilon}_{44} + 8\tilde{\epsilon}_{45} + 14\tilde{\epsilon}_{46} + 13\tilde{\epsilon}_{47} + 9\tilde{\epsilon}_{48} + 7\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 10\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^3 \\
&- \left(14\tilde{\epsilon}_{52} + 15\tilde{\epsilon}_{53} + 17\tilde{\epsilon}_{54} + 16\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^3 - 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^5 \left(\sigma^{(i-1)}\right)^3 + \tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_5 + \tilde{\delta}_6 + 2\tilde{\delta}_8\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(2\tilde{\delta}_{10} + 2\tilde{\delta}_{11} + 3\tilde{\delta}_{13} + 5\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- \left(4\tilde{\delta}_{15} + 4\tilde{\delta}_{16} + 6\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&- 10\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 3\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 6\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 3\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
&+ 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 3\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{1,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_2 - \tilde{\gamma}_3\right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \left(\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + 2\tilde{\gamma}_4 a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)}\right)^2 + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 3\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{4,B}^{(i-1,i-1)} \\
&- 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned}
\tag{42}
$$

$$
\begin{aligned}
\epsilon_{15,A}^{(i,i-1)} = {} & \epsilon_{15,B}^{(i-1,i-1)} + \tilde{\epsilon}_{15} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(9\tilde{\epsilon}_{29} + 4\tilde{\epsilon}_{31} + \tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} - 3\tilde{\epsilon}_{40} - 2\tilde{\epsilon}_{41} - \tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
& - \left(12\tilde{\epsilon}_{43} + 9\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 12\tilde{\epsilon}_{46} + 9\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} - 3\tilde{\epsilon}_{49} - 5\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
& - \left(14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} + 7\tilde{\epsilon}_{54} + 2\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 - 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_5 + \tilde{\delta}_7 - \tilde{\delta}_8 - 2\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
& - \left(2\tilde{\delta}_{11} + \tilde{\delta}_{12} - 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - \tilde{\delta}_{13} - \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
& - \left(6\tilde{\delta}_{15} + 5\tilde{\delta}_{16} - \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - \left(7\tilde{\delta}_{15} + \tilde{\delta}_{16}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
& + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 8\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + \left(2\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& - 8\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& - \left(4\tilde{\gamma}_4 + 3\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& + 8\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} + 2\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 - \left(4\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
& - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 4\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{12,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} \\
& + \tilde{\beta} a_i \nu^{(i-1)} \delta_{12,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{43}
$$

$$
\begin{aligned}
\epsilon_{16,A}^{(i,i-1)} = {} & \epsilon_{16,B}^{(i-1,i-1)} + \tilde{\epsilon}_{16} a_i^5 \left(\sigma^{(i-1)}\right)^4 + \left(10\tilde{\epsilon}_{29} - 2\tilde{\epsilon}_{30} + 5\tilde{\epsilon}_{31} - \tilde{\epsilon}_{33} - 2\tilde{\epsilon}_{35} + 5\tilde{\epsilon}_{36} - \tilde{\epsilon}_{38} - 12\tilde{\epsilon}_{40} - 8\tilde{\epsilon}_{41} - 4\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
& + \left(13\tilde{\epsilon}_{43} + 10\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 13\tilde{\epsilon}_{46} + 10\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} - 14\tilde{\epsilon}_{49} - 11\tilde{\epsilon}_{50} - 24\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
& + \left(14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} - 7\tilde{\epsilon}_{54} - 18\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 - 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 \\
& - \left(2\tilde{\delta}_5 + 3\tilde{\delta}_7 - 4\tilde{\delta}_8 - 8\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(5\tilde{\delta}_{11} - \tilde{\delta}_{13} - 8\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(5\tilde{\delta}_{10} + 9\tilde{\delta}_{11} - 7\tilde{\delta}_{12} - 10\tilde{\delta}_{13} - 8\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(15\tilde{\delta}_{15} + 8\tilde{\delta}_{16} - 11\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(16\tilde{\delta}_{15} - 7\tilde{\delta}_{16} - 7\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 24\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
& - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 12\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} - 5\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - \tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& + 12\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 8\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + \left(3\tilde{\gamma}_4 + 7\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& + 10\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 12\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
& - 32\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} + 16\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 17\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& + 15\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} - 8\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 \\
& + \left(11\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 8\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
& + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 8\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\alpha} a_i^2 \delta_{10,B}^{(i-1,i-1)} + 5\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{13,B}^{(i-1,i-1)} \\
& + 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{10,B}^{(i-1,i-1)} - 5\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{13,B}^{(i-1,i-1)} - 16\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 11\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{44}
$$

$$
\begin{aligned}
\epsilon_{17,A}^{(i,i-1)} &= \epsilon_{17,B}^{(i-1,i-1)} + \tilde{\epsilon}_{17} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(6\tilde{\epsilon}_{29} + 3\tilde{\epsilon}_{31} + \tilde{\epsilon}_{34} + 3\tilde{\epsilon}_{36} + \tilde{\epsilon}_{39} - 9\tilde{\epsilon}_{40} - 6\tilde{\epsilon}_{41} - 3\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(9\tilde{\epsilon}_{43} + 6\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 9\tilde{\epsilon}_{46} + 6\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} - 9\tilde{\epsilon}_{49} - 9\tilde{\epsilon}_{50} - 18\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
&- \left(14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} - 6\tilde{\epsilon}_{54} - 15\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 + 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 \\
&+ \left(\tilde{\delta}_5 + \tilde{\delta}_6 + 3\tilde{\delta}_7 - 3\tilde{\delta}_8 - 6\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{11} - 5\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(5\tilde{\delta}_{10} + 5\tilde{\delta}_{11} - 9\tilde{\delta}_{12} - 8\tilde{\delta}_{13} - 5\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(9\tilde{\delta}_{15} + 6\tilde{\delta}_{16} - 7\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(10\tilde{\delta}_{15} - 10\tilde{\delta}_{16} - 7\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + 3\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 6\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_4 + 7\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - 3\left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} - 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_2 - 6\tilde{\gamma}_3\right) a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&- 8\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \left(2\tilde{\gamma}_4 - 5\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 3\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{14,B}^{(i-1,i-1)} \\
&+ 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{14,B}^{(i-1,i-1)} + 13\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 9\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{45}
$$

$$
\begin{aligned}
\epsilon_{18,A}^{(i,i-1)} &= \epsilon_{18,B}^{(i-1,i-1)} + \tilde{\epsilon}_{18} a_i^5 \left(\sigma^{(i-1)}\right)^4 + \left(6\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} - 3\tilde{\epsilon}_{32} - \tilde{\epsilon}_{33} + 3\tilde{\epsilon}_{36} + 8\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&+ \left(4\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + 9\tilde{\epsilon}_{46} + 6\tilde{\epsilon}_{47} + 3\tilde{\epsilon}_{48} + 8\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 15\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
&+ \left(4\tilde{\epsilon}_{52} + 9\tilde{\epsilon}_{53} + 19\tilde{\epsilon}_{54} + 22\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 + 25\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_6 - 3\tilde{\delta}_8 - 6\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- \left(2\tilde{\delta}_{10} + \tilde{\delta}_{11} - 6\tilde{\delta}_{12} - 2\tilde{\delta}_{13} - 6\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{10} - \tilde{\delta}_{11} - 15\tilde{\delta}_{12} - 9\tilde{\delta}_{13} - 5\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- \left(5\tilde{\delta}_{15} - 10\tilde{\delta}_{16} - 17\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{15} + 18\tilde{\delta}_{16} + 13\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&+ 35\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} \\
&- 3\left(\tilde{\gamma}_2 - 2\tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 3\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 6\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 6\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 3\left(2\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 3\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&- 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} + 6\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&+ \left(3\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 11\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 + 3\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{5,B}^{(i-1,i-1)} \\
&- 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + 12\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - 12\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&- 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{46}
$$

$$
\begin{aligned}
\epsilon_{19,A}^{(i,i-1)} &= \epsilon_{19,B}^{(i-1,i-1)} + \tilde{\epsilon}_{19} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(12\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} + 4\tilde{\epsilon}_{31} - 3\tilde{\epsilon}_{32} + \tilde{\epsilon}_{34} - 2\tilde{\epsilon}_{35} + 5\tilde{\epsilon}_{36} + 6\tilde{\epsilon}_{40} + 5\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(12\tilde{\epsilon}_{43} + 9\tilde{\epsilon}_{44} + 4\tilde{\epsilon}_{45} + 13\tilde{\epsilon}_{46} + 12\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} + 5\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 9\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
&- \left(12\tilde{\epsilon}_{52} + 13\tilde{\epsilon}_{53} + 22\tilde{\epsilon}_{54} + 23\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 - 26\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 \\
&- \left(2\tilde{\delta}_5 + 5\tilde{\delta}_6 - 2\tilde{\delta}_8 - 4\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(4\tilde{\delta}_{10} + \tilde{\delta}_{11} - 6\tilde{\delta}_{12} - 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ \left(7\tilde{\delta}_{10} + \tilde{\delta}_{11} - 17\tilde{\delta}_{12} - 4\tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(7\tilde{\delta}_{15} - 8\tilde{\delta}_{16} - 9\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ \left(5\tilde{\delta}_{15} - 15\tilde{\delta}_{16} - 7\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 15\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 12\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + \left(5\tilde{\gamma}_2 - 4\tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&- 12\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 3\left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
&- \left(10\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 12\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
&- 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} + 2\left(\tilde{\gamma}_2 - 2\tilde{\gamma}_3\right) a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&- \left(10\tilde{\gamma}_4 + 6\tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\left(2\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&- 22\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 2\tilde{\alpha} a_i^2 \delta_{10,B}^{(i-1,i-1)} - 5\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{10,B}^{(i-1,i-1)} \\
&+ 5\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - 10\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&+ 10\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{47}
$$

$$
\begin{aligned}
\epsilon_{20,A}^{(i,i-1)} &= \epsilon_{20,B}^{(i-1,i-1)} + \tilde{\epsilon}_{20} a_i^5 \left(\sigma^{(i-1)}\right)^4 + \left(8\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} + 3\tilde{\epsilon}_{31} - \tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{36} - 6\tilde{\epsilon}_{40} - 4\tilde{\epsilon}_{41} - 3\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&+ \left(10\tilde{\epsilon}_{43} + 7\tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 12\tilde{\epsilon}_{46} + 8\tilde{\epsilon}_{47} + 4\tilde{\epsilon}_{48} - 7\tilde{\epsilon}_{49} - 5\tilde{\epsilon}_{50} - 13\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
&+ \left(10\tilde{\epsilon}_{52} + 12\tilde{\epsilon}_{53} - 7\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 - 4\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_6 - \tilde{\delta}_7 + 2\tilde{\delta}_8 + 4\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&+ 2\left(\tilde{\delta}_{12} - 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} + \left(2\tilde{\delta}_{11} - 4\tilde{\delta}_{12} - 6\tilde{\delta}_{13} - 5\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&+ 2\left(\tilde{\delta}_{16} - 3\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} + 3\left(\tilde{\delta}_{15} - 2\tilde{\delta}_{16} - 2\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 7\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 8\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} \\
&- \left(4\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 8\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + \left(3\tilde{\gamma}_4 + 4\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
&+ \left(8\tilde{\gamma}_4 + 5\tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} - 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&+ 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 11\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&- 8\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} - 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + 2\left(3\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&+ 10\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{7,B}^{(i-1,i-1)} \\
&- 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{48}
$$

$$
\begin{aligned}
\epsilon_{21,A}^{(i,i-1)} &= \epsilon_{21,B}^{(i-1,i-1)} + \tilde{\epsilon}_{21} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(2\tilde{\epsilon}_{29} + \tilde{\epsilon}_{31} + \tilde{\epsilon}_{33} + 2\tilde{\epsilon}_{34} + \tilde{\epsilon}_{36}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(3\tilde{\epsilon}_{43} + 3\tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{50}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 - 3\left(2\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&- 6\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_7 + \tilde{\delta}_8\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(2\tilde{\delta}_{10} + \tilde{\delta}_{11}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(5\tilde{\delta}_{15} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(2\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} \\
&- \left(3\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \sigma^{(i-1)} \delta_{8,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{49}
$$

$$
\begin{aligned}
\epsilon_{22,A}^{(i,i-1)} &= \epsilon_{22,B}^{(i-1,i-1)} + \tilde{\epsilon}_{22} a_i^5 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} - 2\tilde{\epsilon}_{32} - \tilde{\epsilon}_{33} - \tilde{\epsilon}_{35} + 3\tilde{\epsilon}_{40} + 2\tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} - \tilde{\epsilon}_{47} - 2\tilde{\epsilon}_{49} - 4\tilde{\epsilon}_{50} - 4\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 - \left(2\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} - 4\tilde{\epsilon}_{54} - 6\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&+ 5\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_5 + \tilde{\delta}_6 - 2\tilde{\delta}_7 - \tilde{\delta}_8 - \tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(\tilde{\delta}_{10} - 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ \left(4\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(\tilde{\delta}_{15} - \tilde{\delta}_{16} - 4\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ 3\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 9\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} + 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \left(3\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 7\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&- 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&- \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_2 - \tilde{\gamma}_3\right) a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + \left(5\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 2\tilde{\gamma}_4 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&+ 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - \tilde{\alpha} a_i^2 \delta_{10,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \sigma^{(i-1)} \delta_{9,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{10,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{50}
$$

$$
\begin{aligned}
\epsilon_{23,A}^{(i,i-1)} &= \epsilon_{23,B}^{(i-1,i-1)} + \tilde{\epsilon}_{23} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(\tilde{\epsilon}_{35} + 3\tilde{\epsilon}_{36} + 3\tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + 7\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(5\tilde{\epsilon}_{46} + 5\tilde{\epsilon}_{47} + 3\tilde{\epsilon}_{48} + 7\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 14\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 - \left(5\tilde{\epsilon}_{53} + 15\tilde{\epsilon}_{54} + 18\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&- 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(2\tilde{\delta}_8 + 5\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 2\left(2\tilde{\delta}_{12} + \tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(10\tilde{\delta}_{12} + 7\tilde{\delta}_{13} + 4\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - \left(8\tilde{\delta}_{16} + 12\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(12\tilde{\delta}_{16} + 9\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 26\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- 11\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&+ 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
&+ 6\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{51}
$$

$$
\begin{aligned}
\epsilon_{24,A}^{(i,i-1)} &= \epsilon_{24,B}^{(i-1,i-1)} + \tilde{\epsilon}_{24} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(\tilde{\epsilon}_{35} - \tilde{\epsilon}_{36} - 3\tilde{\epsilon}_{37} + \tilde{\epsilon}_{39} - 9\tilde{\epsilon}_{40} - 5\tilde{\epsilon}_{41} - 3\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&+ \left(\tilde{\epsilon}_{46} + 3\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + 8\tilde{\epsilon}_{49} + 13\tilde{\epsilon}_{50} + 18\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\epsilon}_{53} + 16\tilde{\epsilon}_{54} + 21\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&+ 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 - 3\left(\tilde{\delta}_8 + 3\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(6\tilde{\delta}_{12} + 2\tilde{\delta}_{13} + 4\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ \left(18\tilde{\delta}_{12} + 12\tilde{\delta}_{13} + 6\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + \left(10\tilde{\delta}_{16} + 14\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ \left(21\tilde{\delta}_{16} + 15\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 30\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ 18\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 5\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + \tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{6,B}^{(i-1,i-1)} + 12\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&+ 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{52}
$$

$$
\begin{aligned}
\epsilon_{25,A}^{(i,i-1)} &= \epsilon_{25,B}^{(i-1,i-1)} + \tilde{\epsilon}_{25} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(\tilde{\epsilon}_{35} + \tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} + \tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(2\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} + \tilde{\epsilon}_{50} + 2\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 - 2\left(\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&- 3\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{12} a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- 2\left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 4\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- 3\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 8\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{7,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{53}
$$

$$
\begin{aligned}
\epsilon_{26,A}^{(i,i-1)} &= \epsilon_{26,B}^{(i-1,i-1)} + \tilde{\epsilon}_{26} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} - 6\tilde{\epsilon}_{40} - 4\tilde{\epsilon}_{41} - 2\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(\tilde{\epsilon}_{47} + 2\tilde{\epsilon}_{48} - 6\tilde{\epsilon}_{49} - 8\tilde{\epsilon}_{50} - 14\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 - \left(3\tilde{\epsilon}_{53} - 10\tilde{\epsilon}_{54} - 15\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&+ 13\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 - 4\left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_{13} - 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ 8\left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 4\left(\tilde{\delta}_{16} - 2\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ 12\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} + 22\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&+ 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - 6\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&- 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} + 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 + 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{8,B}^{(i-1,i-1)} + 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{54}
$$

$$
\begin{aligned}
\epsilon_{27,A}^{(i,i-1)} &= \epsilon_{27,B}^{(i-1,i-1)} + \tilde{\epsilon}_{27} a_i^5 \left(\sigma^{(i-1)}\right)^4 - \left(2\tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + 9\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \\
&- \left(2\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + 9\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 17\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 - \left(\tilde{\epsilon}_{53} + 15\tilde{\epsilon}_{54} + 20\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 \\
&- 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_8 + 4\tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{14} a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(8\tilde{\delta}_{12} + 5\tilde{\delta}_{13} + 2\tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 8\tilde{\delta}_{17} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- \left(9\tilde{\delta}_{16} + 6\tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 20\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 7\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&+ 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 \\
&+ 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{9,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{55}
$$

$$
\begin{aligned}
\epsilon_{28,A}^{(i,i-1)} &= \epsilon_{28,B}^{(i-1,i-1)} + \tilde{\epsilon}_{28} a_i^5 \left(\sigma^{(i-1)}\right)^4 - 2\left(\tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}\right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 - \left(3\tilde{\epsilon}_{49} + 3\tilde{\epsilon}_{50} + 5\tilde{\epsilon}_{51}\right) a_i^3 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^4 \\
&- \left(4\tilde{\epsilon}_{54} + 6\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^4 - 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^4 \left(\sigma^{(i-1)}\right)^4 + \left(\tilde{\delta}_8 + \tilde{\delta}_9\right) a_i^4 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- 2\tilde{\delta}_{14} a_i^3 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 2\left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{17} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} \\
&- 3\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} - 10\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \alpha_B^{(i-1,i-1)} - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^2 \beta_B^{(i-1,i-1)} \\
&- \tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)}\right)^2 + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{3,B}^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\alpha_B^{(i-1,i-1)}\right)^2 - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{56}
$$

$$
\begin{aligned}
\epsilon_{29,A}^{(i,i-1)} &= \epsilon_{29,B}^{(i-1,i-1)} + \tilde{\epsilon}_{29} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(3\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(7\tilde{\epsilon}_{52} + 7\tilde{\epsilon}_{53} + 3\tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 + 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 + \left(\tilde{\delta}_{10} + \tilde{\delta}_{11}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + \left(4\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&- \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 3\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{16,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} \\
&- 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{16,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{57}
$$

$$
\begin{aligned}
\epsilon_{30,A}^{(i,i-1)} &= \epsilon_{30,B}^{(i-1,i-1)} + \tilde{\epsilon}_{30} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(4\tilde{\epsilon}_{43} - \tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 3\tilde{\epsilon}_{48} + \tilde{\epsilon}_{49}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(19\tilde{\epsilon}_{52} + 18\tilde{\epsilon}_{53} + 4\tilde{\epsilon}_{54} - \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 + 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 + \left(5\tilde{\delta}_{10} + \tilde{\delta}_{11}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + \left(8\tilde{\delta}_{15} - 3\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ 14\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 8\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - 3\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- 3\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - \left(3\tilde{\gamma}_4 + 4\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 3\left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- 9\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 11\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- 6\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 5\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 3\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
&+ 3\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{10,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{58}
$$

$$
\begin{aligned}
\epsilon_{31,A}^{(i,i-1)} &= \epsilon_{31,B}^{(i-1,i-1)} + \tilde{\epsilon}_{31} a_i^4 \left(\sigma^{(i-1)}\right)^5 - \left(3\tilde{\epsilon}_{43} - 2\tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + 6\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{48} - \tilde{\epsilon}_{50}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&- \left(13\tilde{\epsilon}_{52} + 16\tilde{\epsilon}_{53} + 3\tilde{\epsilon}_{54} - 2\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 - 13\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 - \left(4\tilde{\delta}_{10} + 3\tilde{\delta}_{11}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- 4\left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - \left(13\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- 20\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - 13\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ \left(5\tilde{\gamma}_4 + 4\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6\left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ 13\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ 4\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 6\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
&- 2\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&- 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{59}
$$

$$
\begin{aligned}
\epsilon_{32,A}^{(i,i-1)} &= \epsilon_{32,B}^{(i-1,i-1)} + \tilde{\epsilon}_{32} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} + \tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 + \left(3\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 \\
&+ 3\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 + \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 5\tilde{\delta}_{15} a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ 2\left(2\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 6\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&+ 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + \tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{12,B}^{(i-1,i-1)} \\
&- \tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{60}
$$

$$
\begin{aligned}
\epsilon_{33,A}^{(i,i-1)} &= \epsilon_{33,B}^{(i-1,i-1)} + \tilde{\epsilon}_{33} a_i^4 \left(\sigma^{(i-1)}\right)^5 - \left(3\tilde{\epsilon}_{43} - 2\tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 3\tilde{\epsilon}_{48}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&- \left(19\tilde{\epsilon}_{52} + 18\tilde{\epsilon}_{53} + 3\tilde{\epsilon}_{54} - 2\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 - 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 - \left(9\tilde{\delta}_{10} + 3\tilde{\delta}_{11} - \tilde{\delta}_{13}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- \left(11\tilde{\delta}_{15} + 3\tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - \left(18\tilde{\delta}_{15} - 5\tilde{\delta}_{16} + 3\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- 26\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - 19\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 3\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ 3\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 3\left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&- 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(9\tilde{\gamma}_4 - 2\tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&+ 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 3\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} - 3\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{13,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} \\
&+ 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} - 7\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{61}
$$

$$
\begin{aligned}
\epsilon_{34,A}^{(i,i-1)} &= \epsilon_{34,B}^{(i-1,i-1)} + \tilde{\epsilon}_{34} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(3\tilde{\epsilon}_{43} + 4\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{48}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 + \left(18\tilde{\epsilon}_{52} + 13\tilde{\epsilon}_{53} + 3\tilde{\epsilon}_{54}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 \\
&+ 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 + \left(5\tilde{\delta}_{10} + 2\tilde{\delta}_{11} + \tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + \left(9\tilde{\delta}_{15} + 2\tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ \left(11\tilde{\delta}_{15} - \tilde{\delta}_{16} + 4\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 19\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- 3\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&- 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 7\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&- 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 3\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} + 2\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{14,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} \\
&- 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{62}
$$

$$
\begin{aligned}
\epsilon_{35,A}^{(i,i-1)} &= \epsilon_{35,B}^{(i-1,i-1)} + \tilde{\epsilon}_{35} a_i^4 \left(\sigma^{(i-1)}\right)^5 - \left(2\tilde{\epsilon}_{43} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} - \tilde{\epsilon}_{49} - 2\tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&- \left(7\tilde{\epsilon}_{52} + 6\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} - \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 - 5\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 \\
&- \left(2\tilde{\delta}_{10} + \tilde{\delta}_{11} + 2\tilde{\delta}_{12} - \tilde{\delta}_{13} - \tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - \left(2\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&- \left(5\tilde{\delta}_{15} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + \left(2\tilde{\gamma}_4 + 3\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&- 2\left(2\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(3\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{10,B}^{(i-1,i-1)} \\
&+ 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{63}
$$

$$
\begin{aligned}
\epsilon_{36,A}^{(i,i-1)} &= \epsilon_{36,B}^{(i-1,i-1)} + \tilde{\epsilon}_{36} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(3\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{50}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 + 3\left(\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 \\
&+ 6\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 + \tilde{\delta}_{12} a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 2\left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 8\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ 4\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
&- 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{64}
$$

$$
\begin{aligned}
\epsilon_{37,A}^{(i,i-1)} &= \epsilon_{37,B}^{(i-1,i-1)} + \tilde{\epsilon}_{37} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(4\tilde{\epsilon}_{43} + 2\tilde{\epsilon}_{45} + 4\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + 3\tilde{\epsilon}_{48} + \tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(14\tilde{\epsilon}_{52} + 16\tilde{\epsilon}_{53} + 4\tilde{\epsilon}_{54} - 3\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 + 4\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(4\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - 2\tilde{\delta}_{12} + \tilde{\delta}_{13}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 4\left(\tilde{\delta}_{15} + \tilde{\delta}_{16}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 10\tilde{\delta}_{15} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - 4\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + 3\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&- 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
&- 14\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- 2\left(3\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 20\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
&+ 2\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{12,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{65}
$$

$$
\begin{aligned}
\epsilon_{38,A}^{(i,i-1)} &= \epsilon_{38,B}^{(i-1,i-1)} + \tilde{\epsilon}_{38} a_i^4 \left(\sigma^{(i-1)}\right)^5 + 2\left(2\tilde{\epsilon}_{43} + \tilde{\epsilon}_{45} + 2\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} - 2\tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(14\tilde{\epsilon}_{52} + 13\tilde{\epsilon}_{53} + 4\tilde{\epsilon}_{54} + 3\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 + 26\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 \\
&+ 2\left(2\tilde{\delta}_{10} + \tilde{\delta}_{11} - 3\tilde{\delta}_{13} - \tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 2\left(2\tilde{\delta}_{15} - 4\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ 2\left(5\tilde{\delta}_{15} - 6\tilde{\delta}_{16} - \tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 22\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 6\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- 4\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} + 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&- 20\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 6\left(\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&+ 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 4\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} + 2\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} \\
&- 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{13,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{66}
$$

$$
\begin{aligned}
\epsilon_{39,A}^{(i,i-1)} &= \epsilon_{39,B}^{(i-1,i-1)} + \tilde{\epsilon}_{39} a_i^4 \left(\sigma^{(i-1)}\right)^5 - \left(6\tilde{\epsilon}_{43} + 3\tilde{\epsilon}_{45} + 6\tilde{\epsilon}_{46} + \tilde{\epsilon}_{48}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 - \left(21\tilde{\epsilon}_{52} + 16\tilde{\epsilon}_{53} + 6\tilde{\epsilon}_{54} + 3\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 \\
&- 25\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 - \left(6\tilde{\delta}_{10} + 3\tilde{\delta}_{11} - \tilde{\delta}_{13} + \tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - 3\left(2\tilde{\delta}_{15} - 2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&- \left(15\tilde{\delta}_{15} - 2\tilde{\delta}_{16} + 5\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - 27\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&- 21\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 6\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 3\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 6\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 12\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} \\
&+ 9\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 18\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ \left(9\tilde{\gamma}_4 - \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 6\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
&- 3\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{14,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&- 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{67}
$$

$$
\begin{aligned}
\epsilon_{40,A}^{(i,i-1)} &= \epsilon_{40,B}^{(i-1,i-1)} + \tilde{\epsilon}_{40} a_i^4 \left(\sigma^{(i-1)}\right)^5 - \left(6\tilde{\epsilon}_{43} + 3\tilde{\epsilon}_{45} + 6\tilde{\epsilon}_{46} + 3\tilde{\epsilon}_{48} - \tilde{\epsilon}_{49} - 3\tilde{\epsilon}_{50} - 3\tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&- \left(21\tilde{\epsilon}_{52} + 21\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} - 8\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 - 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 \\
&- \left(6\tilde{\delta}_{10} + 3\tilde{\delta}_{11} - 3\tilde{\delta}_{12} - \tilde{\delta}_{13} - \tilde{\delta}_{14}\right) a_i^3 \beta_B^{(i-1,i-1)} \left(\sigma^{(i-1)}\right)^3 - \left(6\tilde{\delta}_{15} - \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&- \left(15\tilde{\delta}_{15} - 5\tilde{\delta}_{16} - 2\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ 6\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 3\left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + \left(6\tilde{\gamma}_4 + 7\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&- 12\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 20\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(9\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&+ 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 11\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 6\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} - 3\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} \\
&+ 6\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 9\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{68}
$$

$$
\begin{aligned}
\epsilon_{41,A}^{(i,i-1)} &= \epsilon_{41,B}^{(i-1,i-1)} + \tilde{\epsilon}_{41} a_i^4 \left(\sigma^{(i-1)}\right)^5 + \left(8\tilde{\epsilon}_{43} + 4\tilde{\epsilon}_{45} + 8\tilde{\epsilon}_{46} + 4\tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} - \tilde{\epsilon}_{50} + \tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(28\tilde{\epsilon}_{52} + 28\tilde{\epsilon}_{53} + 7\tilde{\epsilon}_{54} - 4\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 + 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 \\
&+ \left(8\tilde{\delta}_{10} + 4\tilde{\delta}_{11} - 3\tilde{\delta}_{12} - 2\tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 2\left(4\tilde{\delta}_{15} - \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ \left(20\tilde{\delta}_{15} - 3\tilde{\delta}_{16} - 3\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} + 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&- 8\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \left(4\tilde{\gamma}_4 + 7\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 8\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 16\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{2,B}^{(i-1,i-1)} - 20\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
&- 18\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} + 24\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&- 3\left(4\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 30\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 18\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 8\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
&+ 4\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 10\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&+ 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 8\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{69}
$$

$$
\begin{aligned}
\epsilon_{42,A}^{(i,i-1)} &= \epsilon_{42,B}^{(i-1,i-1)} + \tilde{\epsilon}_{42} a_i^4 \left(\sigma^{(i-1)}\right)^5 - \left(2\tilde{\epsilon}_{43} + \tilde{\epsilon}_{45} + 2\tilde{\epsilon}_{46} + \tilde{\epsilon}_{48} - \tilde{\epsilon}_{49} - \tilde{\epsilon}_{50} - \tilde{\epsilon}_{51}\right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^5 \\
&- \left(7\tilde{\epsilon}_{52} + 7\tilde{\epsilon}_{53} - 5\tilde{\epsilon}_{55}\right) a_i^2 \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^5 + 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^3 \left(\sigma^{(i-1)}\right)^5 - \left(2\tilde{\delta}_{10} + \tilde{\delta}_{11} - \tilde{\delta}_{12} - \tilde{\delta}_{14}\right) a_i^3 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} \\
&+ \left(3\tilde{\delta}_{17} - 2\tilde{\delta}_{15}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} - \left(5\tilde{\delta}_{15} - \tilde{\delta}_{16} - 2\tilde{\delta}_{17}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 13\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \alpha_B^{(i-1,i-1)} \\
&+ 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^3 \beta_B^{(i-1,i-1)} + 2\left(\tilde{\gamma}_2 + \tilde{\gamma}_3\right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 2\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 4\left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
&+ 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)}\right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
&+ \left(3\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
&- \tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
&- 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{70}
$$

$$
\begin{aligned}
\epsilon_{43,A}^{(i,i-1)} &= \epsilon_{43,B}^{(i-1,i-1)} + \tilde{\epsilon}_{43} a_i^3 \left(\sigma^{(i-1)}\right)^6 + \left(4\tilde{\epsilon}_{52} + 3\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + 2\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 + 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 \\
&+ 2\tilde{\delta}_{15} a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} + 11\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\beta_B^{(i-1,i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^2 + 3\tilde{\alpha} a_i^2 \delta_{18,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \sigma^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{71}
$$

$$
\begin{aligned}
\epsilon_{44,A}^{(i,i-1)} &= \epsilon_{44,B}^{(i-1,i-1)} + \tilde{\epsilon}_{44} a_i^3 \left(\sigma^{(i-1)}\right)^6 - \left(4\tilde{\epsilon}_{52} + 3\tilde{\epsilon}_{53} + 7\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 - 60\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 \\
&- \left(2\tilde{\delta}_{15} - \tilde{\delta}_{16}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - 43\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} - 3\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} \\
&+ 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 14\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 \\
&- 3\tilde{\alpha} a_i^2 \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{16,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{18,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \gamma_{6,B}^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{72}
$$

$$
\begin{aligned}
\epsilon_{45,A}^{(i,i-1)} &= \epsilon_{45,B}^{(i-1,i-1)} + \tilde{\epsilon}_{45} a_i^3 \left(\sigma^{(i-1)}\right)^6 + \left(6\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + 6\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 + 48\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 \\
&- \left(\tilde{\delta}_{15} - \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} + 31\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} + 4\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} \\
&- \left(\tilde{\gamma}_4 + \tilde{\gamma}_5\right) a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
&- 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 + \tilde{\alpha} a_i^2 \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{73}
$$

$$
\begin{aligned}
\epsilon_{46,A}^{(i,i-1)} &= \epsilon_{46,B}^{(i-1,i-1)} + \tilde{\epsilon}_{46} a_i^3 \left(\sigma^{(i-1)}\right)^6 + \left(\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} - 2\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 - 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 + \tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} \\
&- 14\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&+ 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{15,B}^{(i-1,i-1)},
\end{aligned} \tag{74}
$$

$$\epsilon_{47,A}^{(i,i-1)} = \epsilon_{47,B}^{(i-1,i-1)} + \tilde{\epsilon}_{47} a_i^3 \left(\sigma^{(i-1)}\right)^6 - \left(\tilde{\epsilon}_{53} - 15\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 + 102\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 - 2\tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)}$$
$$+ 74\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} \tag{75}$$
$$- 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{16,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},$$

$$\epsilon_{48,A}^{(i,i-1)} = \epsilon_{48,B}^{(i-1,i-1)} + \tilde{\epsilon}_{48} a_i^3 \left(\sigma^{(i-1)}\right)^6 + \left(5\tilde{\epsilon}_{53} - 12\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 - 69\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6$$
$$- \left(2\tilde{\delta}_{16} + \tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - 48\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} - 3\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} \tag{76}$$
$$- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} + 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} + 19\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{17,B}^{(i-1,i-1)},$$

$$\epsilon_{49,A}^{(i,i-1)} = \epsilon_{49,B}^{(i-1,i-1)} + \tilde{\epsilon}_{49} a_i^3 \left(\sigma^{(i-1)}\right)^6 + \left(\tilde{\epsilon}_{54} + 7\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 + 42\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6$$
$$- \left(\tilde{\delta}_{16} + 2\tilde{\delta}_{17}\right) a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} + 20\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} - 9\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)}$$
$$- \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} \tag{77}$$
$$- 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2 + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},$$

$$\epsilon_{50,A}^{(i,i-1)} = \epsilon_{50,B}^{(i-1,i-1)} + \tilde{\epsilon}_{50} a_i^3 \left(\sigma^{(i-1)}\right)^6 + \left(3\tilde{\epsilon}_{54} - 6\tilde{\epsilon}_{55}\right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 - 63\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 + 2\tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)}$$
$$- 50\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} \tag{78}$$
$$+ 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^2 \gamma_{6,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)} + 13\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2,$$

$$\epsilon_{51,A}^{(i,i-1)} = \epsilon_{51,B}^{(i-1,i-1)} + \tilde{\epsilon}_{51} a_i^3 \left(\sigma^{(i-1)}\right)^6 + 3\tilde{\epsilon}_{55} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^6 + 21\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)}\right)^2 \left(\sigma^{(i-1)}\right)^6 + \tilde{\delta}_{17} a_i^2 \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)}$$
$$+ 20\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \alpha_B^{(i-1,i-1)} + 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^4 \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{4,B}^{(i-1,i-1)} \tag{79}$$
$$- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^2 \left(\beta_B^{(i-1,i-1)}\right)^2,$$

$$\epsilon_{52,A}^{(i,i-1)} = \epsilon_{52,B}^{(i-1,i-1)} + \tilde{\epsilon}_{52} a_i^2 \left(\sigma^{(i-1)}\right)^7 + \tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^7 + \tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \beta_B^{(i-1,i-1)}$$
$$+ \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{18,B}^{(i-1,i-1)}, \tag{80}$$

$$\epsilon_{53,A}^{(i,i-1)} = \epsilon_{53,B}^{(i-1,i-1)} + \tilde{\epsilon}_{53} a_i^2 \left(\sigma^{(i-1)}\right)^7 + 6\tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^7 - 3\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \beta_B^{(i-1,i-1)}$$
$$- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{18,B}^{(i-1,i-1)}, \tag{81}$$

$$\epsilon_{54,A}^{(i,i-1)} = \epsilon_{54,B}^{(i-1,i-1)} + \tilde{\epsilon}_{54} a_i^2 \left(\sigma^{(i-1)}\right)^7 - 14\tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^7 + 6\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)}\right)^5 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)}, \tag{82}$$

$$\epsilon_{55,A}^{(i,i-1)} = \epsilon_{55,B}^{(i-1,i-1)} + \tilde{\epsilon}_{55} a_i^2 \left(\sigma^{(i-1)}\right)^7 + 14\tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)}\right)^7 - 5\tilde{\delta}_{18} a_i \beta_B^{(i-1,i-1)} \left(\sigma^{(i-1)}\right)^5 - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)}\right)^3 \gamma_{6,B}^{(i-1,i-1)}, \tag{83}$$

$$\epsilon_{56,A}^{(i,i-1)} = \epsilon_{56,B}^{(i-1,i-1)} + \tilde{\epsilon}_{56} a_i \left(\sigma^{(i-1)}\right)^8. \tag{84}$$